

## Financial Intermediaries and the Cross-Section of Asset Returns

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### ABSTRACT

Financial intermediaries trade frequently in many markets using sophisticated models. Their marginal value of wealth should therefore provide a more informative stochastic discount factor (SDF) than that of a representative consumer. Guided by theory, we use shocks to the leverage of securities broker-dealers to construct an intermediary SDF. Intuitively, deteriorating funding conditions are associated with deleveraging and high marginal value of wealth. Our *single*-factor model prices size, book-to-market, momentum, and bond portfolios with an  $R^2$  of 77% and an average annual pricing error of 1%—performing as well as standard multifactor benchmarks designed to price these assets.

MODERN FINANCE THEORY asserts that asset prices are determined by their covariances with the stochastic discount factor (SDF), which is usually linked to the marginal value of aggregate wealth. Assets that are expected to pay off in future states with high marginal value of wealth are worth more today, as dictated by investors' first-order conditions. Following this theory, much of the empirical asset pricing literature centers on measuring the marginal value of wealth of a representative investor, typically the average household. Specifically, the SDF is represented by the marginal value of wealth aggregated over all households. However, the logic that leads to this SDF relies on strong assumptions: all households must participate in all markets, there cannot be transactions costs, households have to be able to execute complicated trading

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strategies, the moments of asset returns have to be known, and investment strategies have to be continuously optimized based on forward-looking expectations. If these assumptions are violated for some agents, it can no longer be assumed that the marginal value of wealth of the representative household prices all assets.<sup>1</sup> For example, if some investors trade only in (say) value stocks, their marginal value of wealth can only be expected to correctly price those stocks. In contrast, should there exist a single class of investors that fits the assumptions of modern finance theory, their marginal value of wealth can be expected to price all assets.

This paper shifts attention from measuring the SDF of the average household to measuring a “financial intermediary SDF.” This approach takes us to a new place in the field of empirical asset pricing—rather than emphasize average household behavior, the assumptions of frictionless markets and intertemporally optimizing behavior suggest that financial intermediaries be elevated to the center stage of asset pricing. Indeed, financial intermediaries fit the assumptions of modern finance theory nicely: they trade in many asset classes following often complex investment strategies; they face low transaction costs, which allows trading at high frequencies; and they use sophisticated, continuously updated models and extensive data to form forward-looking expectations of asset returns. Therefore, if we can measure the marginal value of wealth for these active investors, we can expect to price a broad class of assets.<sup>2</sup> In other words, the marginal value of wealth of intermediaries can be expected to provide a more informative SDF.

Backed by recent theories that give financial intermediaries a central role in asset pricing, we argue that the leverage of security broker-dealers is a good empirical proxy for the marginal value of wealth of financial intermediaries and that it can thus be used as a representation of the intermediary SDF. Intuitively, when funding conditions tighten and intermediaries are forced to deleverage, their marginal value of wealth should be high. We find remarkably strong empirical support for this hypothesis: exposures to the broker-dealer leverage factor *alone* can explain the average excess returns on a wide variety of test assets, including equity portfolios sorted by size, book-to-market, and momentum, as well as the cross-section of Treasury bond portfolios sorted by maturity. The broker-dealer leverage factor is successful across all cross-sections in terms of high adjusted  $R^2$ s, low cross-sectional pricing errors, and prices of risk that are significant and remarkably consistent across portfolios.<sup>3</sup> When taking all these criteria into account, our single factor outperforms standard multifactor models tailored to price these cross-sections, including the

<sup>1</sup> See Jagannathan and Wang (2007) for evidence that households may optimize infrequently and Malloy, Moskowitz, and Vissing-Jorgensen (2009) for evidence that limited participation in the stock market can help explain the cross-section of stock returns and the equity premium puzzle.

<sup>2</sup> An insight due to He and Krishnamurthy (2013).

<sup>3</sup> The returns on momentum portfolios have thus far been particularly difficult to connect to risk. We regard the strong pricing performance across transaction-cost-intensive momentum and bond portfolios as an indication that these portfolios are better priced by the SDF of a sophisticated intermediary.



among benchmark portfolio returns. In fact, the mean variance characteristics of the LMP are close to the tangency portfolio on the efficient frontier generated by combinations of the three Fama-French factors and the momentum factor. As a further robustness check, we use the entire cross-section of stock returns to construct portfolios based on our leverage factor betas and find substantial dispersion in average returns that line up well with the postformation leverage betas.

Our empirical results are consistent with a growing theoretical literature on the links between financial institutions and asset prices. Most directly, our results support models in which leverage captures the time-varying balance sheet capacity of financial intermediaries. As funding constraints tighten, balance sheet capacity falls and intermediaries are forced to deleverage by selling assets at fire sale prices, as in the recent financial crisis. These are times when intermediaries' marginal value of wealth is high. Assets that pay off poorly when constraints tighten and leverage falls are therefore risky and must offer high returns. Equivalently, the cross-sectional price of leverage risk should be positive. These theories imply that leverage captures aspects of the intermediary SDF that other measures (such as aggregate consumption growth or the return on the market portfolio) do not capture. A common thread in these theories is the *procyclical* evolution of broker-dealer leverage as leverage increases in good times when funding constraints are relaxed.

We provide empirical support for the view that leverage represents funding constraints by showing that our leverage factor correlates with funding constraint proxies such as volatility, the Baa-Aaa spread, asset growth, and a betting-against-beta factor that goes long leveraged low-beta securities and short high-beta securities. Frazzini and Pedersen (2013) show that investors who face funding constraints prefer to hold naturally high-beta securities rather than leveraging up low-beta ones, resulting in a positive average return spread between a levered low-beta asset and a naturally high-beta asset. This betting-against-beta factor should comove with funding constraints. Consistent with this view, we find that our leverage factor correlates well with the betting-against-beta portfolio and explains the cross-section of returns sorted on betas as well.

To the best of our knowledge, we are the first to conduct cross-sectional asset pricing tests with financial intermediary balance sheet components in the pricing kernel, which provides an explicit link between intermediary balance sheets and asset prices. To quote John H. Cochrane's (2011, p. 1091) Presidential Address on intermediary-based theories of asset pricing, "A crucial question is, as always, what data will this class of theories use to measure discount rates? Arguing over puzzling patterns of prices is weak. The rational-behavioral debate has been doing that for 40 years, rather unproductively. Ideally, one should tie price or discount-rate variation to central items in the models, such as the balance sheets of leveraged intermediaries."

The remainder of the paper is organized as follows. Section I provides a discussion of related literature, reviewing the theoretical rationalizations for the link between financial intermediary leverage and aggregate asset prices.

Section II describes the data and empirical strategy, and Section III conducts a number of asset pricing tests in the cross-section of stock and bond returns. Section IV analyzes the properties of the LMP and forms portfolios sorted on leverage betas, providing a variety of robustness checks. Section V discusses challenges for existing theories and directions for future work. Section VI concludes.

## I. Financial Intermediary Asset Pricing

An intermediary pricing kernel arises if the balance sheet capacity of intermediaries directly impacts asset price dynamics, as is the case in the literature on limits to arbitrage. In such frameworks, the leverage of financial intermediaries measures the tightness of intermediary funding constraints and therefore their marginal value of wealth. As risk constraints—such as those on intermediary funding—tighten, intermediaries are forced to sell assets by deleveraging, as during the financial crisis. Assets that covary strongly with intermediary funding conditions are thus risky, and equilibrium excess returns reflect this risk. More formally, the marginal value of wealth is related to intermediaries' Lagrange multiplier on the funding constraint, making low leverage states ones with high marginal utility. Prominent examples of such theories include Gromb and Vayanos (2002), Brunnermeier and Pedersen (2009), Geanakoplos (2010), and Shleifer and Vishny (1997).

Brunnermeier and Pedersen (2009) show how funding liquidity enters the pricing kernel when investors are risk-neutral and face funding constraints. Specifically, let  $\phi_1$  be the Lagrange multiplier on the time-one margin constraint and let  $W_1$  denote time-one wealth. Risk-neutral investors subject to these constraints maximize  $E_0[\phi_1 W_1]$ . Immediately, we see that the SDF is given by  $\frac{\phi_1}{E_0[\phi_1]}$ , as this problem is clearly equivalent to an investor maximizing the present value of her portfolio using  $\phi_1$  as the time-one state price. Thus, even with risk-neutrality, the constraint gives rise to nontrivial state pricing since it places higher value on states in which funding constraints are tighter. Taking the first-order conditions of the risk-neutral intermediary, the ex-ante time-zero price of security  $j$  is given by

$$p_{0,j} = E_0[p_{1,j}] + \frac{\text{Cov}_0[p_{1,j}, \phi_1]}{E_0[\phi_1]}, \quad (1)$$

where  $\phi_1$  is the Lagrange multiplier on the time-one margin constraint, which is monotonically decreasing in time-one leverage (see Brunnermeier and Pedersen's (2009) equation 31). Rearranging and stating equation (1) in returns, we have the following equation for excess returns:

$$E_0[R_{1,j}^e] = -\frac{\text{Cov}_0[\phi_1, R_{1,j}^e]}{E_0[\phi_1]}. \quad (2)$$

When funding constraints tighten, intermediaries are forced to deleverage by selling off assets they can no longer finance. Since leverage is a proxy for funding conditions in Brunnermeier and Pedersen's (2009) model, they provide justification for our one-factor leverage model. More specifically, because funding constraints at time-one are always binding, leverage directly measures such constraints and hence measures the marginal value of wealth. In the language of Brunnermeier and Pedersen (2009), we propose  $\phi_1 \approx a - b \ln(\text{Leverage}_1)$ , such that lower  $\text{Leverage}_1$  corresponds to tighter funding constraints. We therefore have the approximation

$$E_0 \left[ R_{1,j}^e \right] = \lambda \text{Cov}_0 \left[ \ln(\text{Leverage}_1), R_{1,j}^e \right], \quad (3)$$

where  $\lambda > 0$ . Thus, assets that covary with leverage are risky and earn a larger risk premium.

Along similar lines, Danielsson, Shin, and Zigrand (2009) consider risk-neutral financial intermediaries that are subject to a value at risk (VaR) constraint in a dynamic model.<sup>4</sup> The intermediaries' demand for risky assets depends on the Lagrange multiplier of the VaR constraint that reflects effective risk aversion. In equilibrium, asset prices depend on the level of effective risk aversion and hence on the leverage of the intermediaries—times of low intermediary leverage are times when effective risk aversion is high. As a result, financial intermediary leverage directly enters the equilibrium SDF.

Adrian and Boyarchenko (2012) develop an equilibrium model of the macroeconomy where intermediaries are subject to risk-based funding constraints that give rise to an equilibrium representation with intermediary leverage as the key state variable. Importantly, the equilibrium pricing kernel can be expressed as a function of shocks to financial intermediary leverage, which represent funding conditions, and shocks to output. When intermediaries experience an adverse shock to their funding, their effective risk aversion endogenously increases as their leverage declines. As a result, Adrian and Boyarchenko (2012) predict that the price of risk of intermediary leverage is positive, consistent with our empirical results. It is important to note that leverage—not wealth—is the key measure of the marginal value of wealth in these models.

An alternative approach to modeling an intermediary pricing kernel is proposed by He and Krishnamurthy (2013). In their setup, intermediary wealth, not intermediary leverage, is the key state variable. They argue that financial intermediaries are the marginal investor, and, as a result, the SDF is proportional to the wealth growth of the intermediary sector, giving an intermediary CAPM. Brunnermeier and Sannikov (2014) derive a closely related equilibrium asset pricing model with financial intermediaries where intermediation arises as an outcome of principal-agent problems. Their model also predicts

<sup>4</sup> Other examples include Chabakauri (2010), Prieto (2010), and Rytchkov (2014), which are dynamic versions of models with funding constraints. These theories build on heterogeneous agent extensions of the Intertemporal Capital Asset Pricing Model (ICAPM) of Merton (1973), where leverage arises as a reduced-form representation of relevant state variables, capturing shifts in the marginal value of wealth.

that shocks to intermediary wealth are the relevant measure of systematic risk. However, empirically, we will see that exposure to intermediary wealth shocks, as measured by the total return on the financial sector, does not explain the cross-section of average returns. These results are detailed in the Internet Appendix.<sup>5</sup> In addition, both He and Krishnamurthy (2013) and Brunnermeier and Sannikov (2014) feature countercyclical intermediary leverage, thus predicting a negative price of risk of intermediary leverage.

While many theories with linkages between financial intermediaries and asset pricing have been proposed, the insight that financial institutions' balance sheets contain information about the real economy and expected asset returns has received less empirical attention. Adrian and Shin (2010) document that security broker-dealers adjust their financial leverage aggressively as economic conditions change. Broker-dealers' balance sheet management practices result in highly procyclical leverage. Recently, Adrian, Etula, and Shin (2009), Adrian, Moench, and Shin (2013), and Etula (2013) show that broker-dealer leverage contains strong predictive power for asset prices. Muir (2012) finds that intermediary equity forecasts returns and that risk premia are particularly high around financial crises. The predictive power of intermediary balance sheets for stock and bond returns suggests that they contain valuable information about the evolution of risk premia over time. In this paper, we show that broker-dealer leverage can *price* assets by connecting the cross-section of returns to exposures to broker-dealer leverage shocks.

## II. Data and Empirical Approach

Motivated by the theories on financial intermediaries and aggregate asset prices, we identify shocks to the leverage of security broker-dealers as a proxy for shocks to the pricing kernel. We use the following measure of broker-dealer (BD) leverage:

$$Leverage_t^{BD} = \frac{Total\ Financial\ Assets_t^{BD}}{Total\ Financial\ Assets_t^{BD} - Total\ Liabilities_t^{BD}}. \quad (4)$$

We construct this variable using aggregate quarterly data on the levels of total financial assets and total financial liabilities of security broker-dealers as captured in Table L.129 of the Federal Reserve *Flow of Funds*. Table I provides the breakdown of assets and liabilities of security brokers and dealers as of the end of 2010.

### A. Aggregate Balance Sheet of Broker-Dealers

The balance sheet composition of security brokers and dealers combined with the evidence of Adrian and Shin (2010) on intermediary balance sheet

<sup>5</sup> The Internet Appendix is available in the online version of the article on *The Journal of Finance* website.



**Table I**  
**Assets and Liabilities of Security Brokers and Dealers as of the End of 2010Q4**

This table presents the assets and liabilities from the *Flow of Funds*. The miscellaneous assets and liabilities are items from the Focus reports. The amounts are in billions of dollars, not seasonally adjusted. (Source: U.S. *Flow of Funds* Table L.129., released March 10, 2011.)

| Assets from <i>Flow of Funds</i> (billions)              |                  | Liabilities from <i>Flow of Funds</i> (billions) |                  |
|----------------------------------------------------------|------------------|--------------------------------------------------|------------------|
| Cash (including segregated cash)                         | \$96.9           | Net repo                                         | \$404.7          |
| Credit market instruments                                | \$557.6          | Corporate and foreign bonds                      | \$129.7          |
| Commercial paper                                         | \$36.2           | Trade payables                                   | \$18.1           |
| Treasury securities (net of shorts)                      | \$94.5           | Security credit                                  | \$936.6          |
| Agencies                                                 | \$149.8          | Taxes payable                                    | \$3.6            |
| Municipal securities                                     | \$40.0           | Miscellaneous liabilities*                       | \$480.7          |
| Corporate and foreign bonds                              | \$185.6          | Payables to brokers and dealers                  |                  |
| Other (syndicated loans, etc.)                           | \$51.4           | Securities sold not yet purchased                |                  |
| Corporate equities                                       | \$117.2          | Payables                                         |                  |
| Security credit                                          | \$278.2          | Subordinated liabilities                         |                  |
| Miscellaneous assets*                                    | \$1,025.3        |                                                  |                  |
| Receivables                                              |                  |                                                  |                  |
| Reverse repos                                            |                  |                                                  |                  |
| Options and arbitrage                                    |                  |                                                  |                  |
| Commodities                                              |                  |                                                  |                  |
| Investments not readily marketable                       |                  |                                                  |                  |
| Securities borrowed                                      |                  |                                                  |                  |
| Secured demand notes                                     |                  |                                                  |                  |
| Membership in exchanges                                  |                  |                                                  |                  |
| Affiliates, subsidiaries,<br>and associated partnerships |                  |                                                  |                  |
| Property, furniture, equipment, etc.                     |                  |                                                  |                  |
| <b>TOTAL</b>                                             | <b>\$2,075.1</b> | <b>TOTAL</b>                                     | <b>\$1,973.4</b> |

\*Subcategories implicit in FOCUS reports.

adjustments suggest that shocks to financial intermediaries' leverage growth may provide a more informative pricing kernel than the growth rate of average consumption or the balance sheet of the average market participant usually used as pricing kernel proxies. The asset side of broker-dealers' balance sheets consists largely of risky assets, while a substantial portion of the liability side consists of short-term, collateralized borrowing (net repos make up roughly 25% to 30% of liabilities). Increases in broker-dealer leverage as captured by the *Flow of Funds* thus correspond primarily to increases in risk-taking. Moreover, since the leverage of broker-dealers computed from the *Flow of Funds* is a net number, we do not emphasize the *level* of broker-dealer leverage but instead focus on innovations to broker-dealer leverage.

The total financial assets of \$2,075 billion in 2010 are divided into five main categories: (1) cash, (2) credit market instruments, (3) equities, (4) security credit, and (5) miscellaneous assets. The *Flow of Funds* further reports finer categories of credit market instruments (commercial paper, Treasury



securities, agencies, municipal securities and loans, corporate and foreign bonds, and syndicated loans). The category “miscellaneous assets” arises as the *Flow of Funds* statistics only keep track of a limited number of asset classes but security broker-dealers are involved in many financial transactions not captured by these broad asset classes. Because the security broker-dealer statistics are derived from the SEC’s FOCUS reports, it is possible to reconstruct the missing items of the miscellaneous assets from those reports. Table I lists the asset categories that comprise miscellaneous assets: receivables;<sup>6</sup> reverse repos; options and arbitrage; spot commodities; investments not readily marketable; securities borrowed under subordination agreements; secured demand notes; membership in exchanges; and investment in and receivables from affiliates, subsidiaries, and associated partnerships. A further category that appears in the FOCUS reports but not in the *Flow of Funds* is nonfinancial assets (property, furniture, etc.). The liabilities that the *Flow of Funds* reports are (1) net repo, (2) corporate and foreign bonds, (3) trade payables, (4) security credit, (5) taxes payable, and (6) miscellaneous liabilities. The miscellaneous liabilities can be extracted from the FOCUS reports: payables,<sup>7</sup> securities sold not yet purchased, and liabilities subordinated to claims of general creditors. The repos that appear on the liability side of the flow of funds are the difference between repos and reverse repos from the FOCUS reports. The *Flow of Funds* thus only reports the net repo funding of broker-dealers, not the total size of the repo market.<sup>8</sup>

### B. Time Series of Broker-Dealer Leverage

While the *Flow of Funds* data begin in 1952Q1, the data from the broker-dealer sector prior to 1968 raise suspicions: broker-dealer equity is *negative* over the period 1952Q1 to 1960Q4 and extremely low for most of the 1960s, resulting in unreasonably high leverage ratios. As a result, we begin our sample in 1968Q1. However, we show that our results do not depend on this exact date and are robust to using a five-year window around this period.

We construct the leverage factor as seasonally adjusted log changes in the level of broker-dealer leverage:

$$LevFac_t = \left[ \Delta \ln \left( Leverage_t^{BD} \right) \right]^{SA}. \quad (5)$$

We seasonally adjust the log changes using quarterly seasonal dummies in real time, which means that we compute an expanding-window regression at each date using the data up to that date. This adjustment ensures that we have

<sup>6</sup> Receivables from broker-dealers and clearing organizations, and reverse repos from broker-dealers, are subtracted from the total assets in the FOCUS reports because the *Flow of Funds* reports the balance sheet for the aggregate broker-dealer sector.

<sup>7</sup> Payables to broker-dealers and clearing corporations are subtracted from the FOCUS report liabilities before entering the *Flow of Funds*.

<sup>8</sup> One peculiarity of the *Flow of Funds* is that Foreign Direct Investment in U.S. broker-dealers is subtracted from total liabilities.

Table II  
Broker-Dealer Leverage Is Procyclical

This table presents the correlation of U.S. broker-dealer leverage growth with a selection of variables, including the log asset growth of U.S. broker-dealers, market volatility (constructed quarterly using weekly data of the value-weighted market return), the Baa-Aaa spread, and the value-weighted stock return of the U.S. financial sector. The sample is 1968Q1 to 2009Q4.

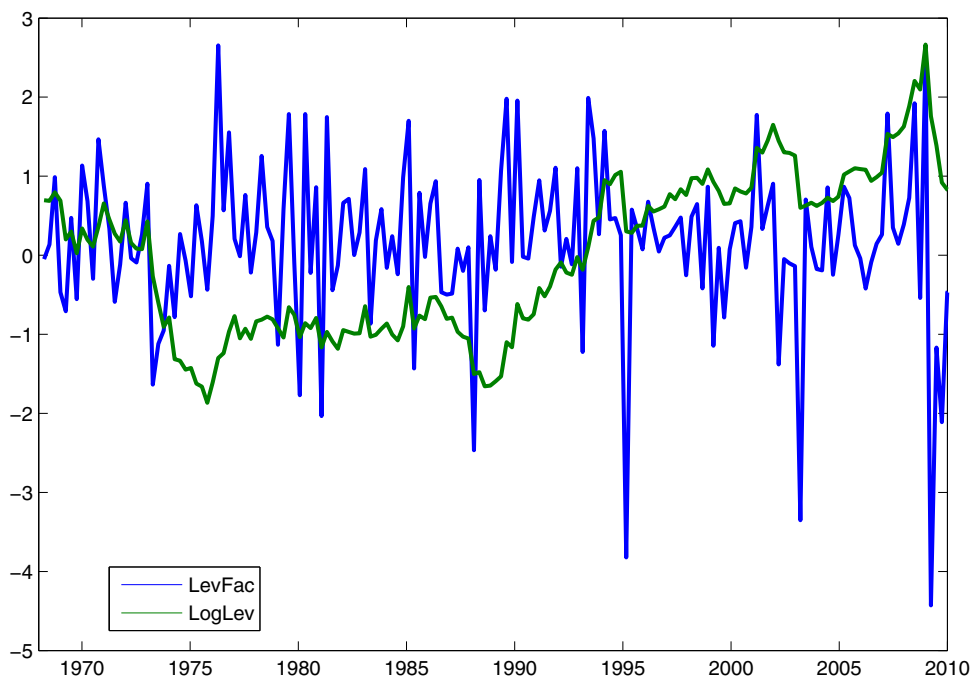
| Correlation of Broker-Dealer Leverage Factor with: |                                   |                      |                   |                            |
|----------------------------------------------------|-----------------------------------|----------------------|-------------------|----------------------------|
|                                                    | Log Broker-Dealer<br>Asset Growth | Market<br>Volatility | Baa-Aaa<br>Spread | Financials<br>Stock Return |
| $\rho$                                             | 0.73                              | −0.37                | −0.16             | 0.18                       |
| $p$ -value                                         | 0.00                              | 0.00                 | 0.03              | 0.02                       |

real-time leverage shocks.<sup>9</sup> The results are robust to using alternate measures, such as more complicated seasonal filtering techniques, but we prefer the current construction for its simplicity. There is strong evidence of seasonal components in the data—in a regression using the full sample, all seasonal dummies are highly statistically significant. Note that, due to the high persistence of the leverage series, using log changes in leverage as shocks is virtually identical to using log innovations from an AR(1) model. Therefore, we prefer to use log changes rather than adding the complication of an AR(1) specification.

Table II documents the correlation of our leverage factor with other intermediary indicators. We confirm the strong correlation between the leverage factor and asset growth (0.73). Leverage shocks are negatively related to volatility (−0.37) and the Baa–Aaa default spread (−0.16), and positively related to the value-weighted return on the financial sector (0.18). Therefore, decreases in leverage are associated with a reduction in broker-dealer assets, spikes in volatility and credit spreads, and decreases in financial sector equity—all of which are consistent with a high marginal value of wealth for intermediaries. These findings are also consistent with the “margin spiral” of Brunnermeier and Pedersen (2009), where both increases in volatility and declines in asset value cause funding conditions to deteriorate, forcing intermediaries to deleverage and reduce their balance sheet size.

We display a plot of broker-dealer leverage and leverage shocks in Figure 2. We see sharp drops in leverage during the 1970s oil crisis, the 1987 stock market crash, and, most notably, in the recent financial crisis (particularly after the Lehman collapse, but we also see large drops in the subsequent quarters). This plot again suggests that many large decreases in broker-dealer leverage are indeed associated with times of macroeconomic and financial sector turmoil, in line with the idea that sharp decreases in leverage represent “bad times” where funding is tight and the marginal value of intermediary wealth is high. However, we also note that we do not explain the magnitude of every drop in the leverage factor. In particular, we observe a large drop in 2003, which was

<sup>9</sup> We initialize the series in 1968Q1 using data from the previous 10 quarters to compute our shocks. However, our results are robust to starting at later dates to allow for a longer initialization (e.g., starting in 1971Q1 and using 22 quarters to initialize the series).



**Figure 2. Log-leverage and leverage factor.** We plot the log-leverage and the leverage factor (changes in log-leverage) of security broker-dealers, 1968Q1 to 2009Q4. We standardize each series to have zero mean and unit variance for convenience.

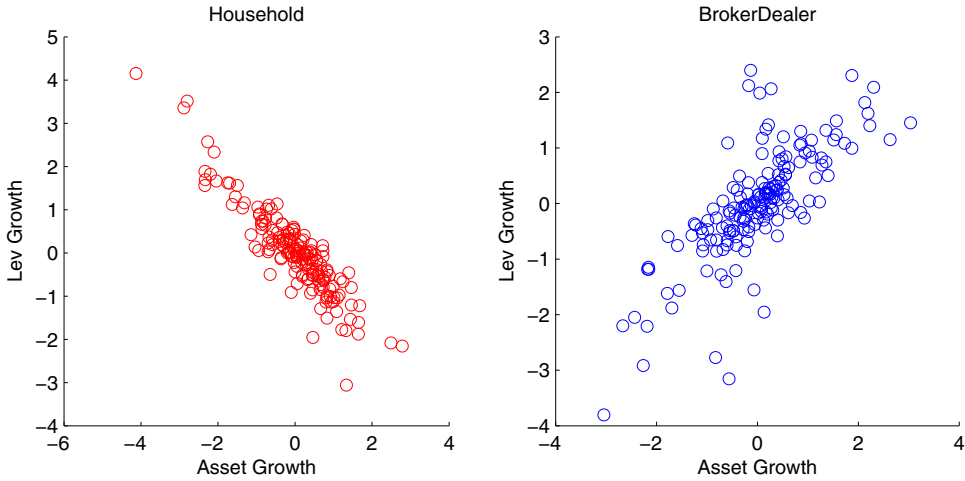
associated with a high volatility period, as well as a large drop in 1994Q4, which occurs during the Mexican Peso Crisis. While it is likely that these periods had an adverse effect on intermediaries, the magnitudes of these drops are large.

We also emphasize the *procyclical* evolution of broker-dealer leverage, which is precisely opposite to the mechanical effects one would expect. To highlight the procyclicality, we plot leverage growth versus asset growth for broker-dealers and contrast it with that of households in Figure 3. If there is no active balance sheet adjustment, we expect the two to be negatively correlated—as asset values improve, leverage mechanically falls as equity grows, and vice versa. This is exactly what we see for households. In contrast, broker-dealers display the exact *opposite* pattern. Asset growth and leverage growth are positively correlated. Increases in asset values are thus associated with increases in leverage, supporting our claim that broker-dealers manage balance sheets aggressively and actively.

### C. Empirical Strategy

We test our leverage factor model in the cross-section of asset returns using a linear factor model. Equivalently, we propose an SDF for excess returns that is affine in the financial intermediary leverage factor:

$$SDF_t = 1 - bLevFac_t.$$



**Figure 3. Cyclicity of leverage.** We plot leverage growth versus asset growth for households and broker-dealers. In both cases, leverage is defined as (Total Assets)/(Total Assets – Total Liabilities). A downward-sloping line (households) indicates passive balance sheet management, as increasing asset values mechanically decrease leverage and vice versa. An upward-sloping line (broker-dealers) indicates active balance sheet management, whereby increases in asset values are associated with increases in leverage. Data are quarterly, 1968 to 2009, from the *Flow of Funds*.

The no-arbitrage condition for asset  $i$ 's return in excess of the risk-free rate is

$$\begin{aligned} 0 &= E[R_{i,t}^e SDF_t] \\ &= E[R_{i,t}^e (1 - b Lev Fac_t)]. \end{aligned}$$

Rearranging and using the definition of covariance, we obtain the factor model,

$$E[R_{i,t}^e] = b \text{Cov}(R_{i,t}^e, Lev Fac_t) \quad (6)$$

$$= \lambda_{Lev} \beta_{i,Lev}, \quad (7)$$

where  $\beta_{i,Lev} = \text{Cov}(R_{i,t}^e, Lev Fac_t) / \text{Var}(Lev Fac_t)$  denotes the exposure of asset  $i$  to broker-dealer leverage shocks and  $\lambda_{Lev}$  is the cross-sectional price of risk associated with leverage shocks.

For each asset  $i = 1, \dots, N$ , we estimate the risk exposures from the time-series regression:

$$R_{i,t}^e = c_i + \beta'_{i,f} \mathbf{f}_t + \epsilon_{i,t}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (8)$$

where  $\mathbf{f}$  represents a vector of risk factors. To estimate the cross-sectional price of risk associated with the factors  $\mathbf{f}$ , we run a cross-sectional regression of time-series average excess returns,  $E[R_t^e]$ , on risk factor exposures,

$$E[R_{i,t}^e] = \mu_{\mathbf{R},i} = a + \beta'_{i,f} \lambda_f + \xi_i, \quad i = 1, \dots, N. \quad (9)$$

This approach yields estimates of the cross-sectional prices of risk  $\lambda$  and the average cross-sectional pricing error or zero-beta rate,  $a$ . A good pricing model features an economically small and statistically insignificant intercept ( $a$ ), statistically significant and stable prices of risk ( $\lambda$ ) across different cross-sections of test assets, and individual pricing errors ( $\xi_i$ ) that are close to zero. We measure the size of the pricing errors in several ways: by the cross-sectional adjusted  $R^2$ , which focuses on whether the sum of squared errors is relatively small ( $1 - \sigma_\xi^2 / \sigma_{\mu_{\mathbf{R}}}^2$ ), by the mean absolute pricing error or MAPE ( $\frac{1}{N} \sum |\xi_i|$ ), which focuses less on outliers than the  $R^2$ , and by a  $\chi^2$  statistic that tests whether the pricing errors are jointly zero—measured by a weighted sum of squared pricing errors  $(\xi' \text{cov}(\xi)^{-1} \xi \sim \chi_{N-K}^2$ , where  $K$  is the number of factors and  $\text{cov}(\xi)$  includes the estimation error in  $\beta$ s).<sup>10,11</sup> The latter is the only formal statistical measure of whether the pricing errors are “too big,” while the MAPE and  $R^2$  are easier diagnostics to interpret from an economic standpoint. To correct the standard errors for the preestimation of betas, we report the  $t$ -statistics of Shanken (1992) in addition to the  $t$ -statistics of Fama and MacBeth (1973).

We also provide confidence intervals for the  $R^2$  statistics using bootstrapping as the sample  $R^2$  can be misleading or uninformative due to large sampling errors. We follow Lewellen, Nagel, and Shanken (2010) in computing confidence intervals and relegate the exact details to their paper. The issue with the sample  $R^2$  is that, even if the “true”  $R^2$  is close to zero, the sample  $R^2$  can be fairly large. Similarly, even if the true  $R^2$  is close to one, the sample  $R^2$  will likely be well below one. Therefore, a particular sample  $R^2$  can, in principle, correspond to a large range of true  $R^2$  values. We construct the sampling error for any true  $R^2$  by simulating a model with the true value of the  $R^2$ . We then compute the sampling error via bootstrapping to see what range of sample  $R^2$  could, in principle, correspond to the given true value. We step over all true values from zero to one. We are then able to determine, for any given sample  $R^2$ , the range of true  $R^2$  statistics that is likely to produce the sample value. This range forms our confidence interval.

By following the above evaluation criteria, and applying our single-factor model to a wide range of test assets, we address the criticisms of traditional asset pricing tests raised by Lewellen, Nagel, and Shanken (2010). First, since we use a one-factor model, we avoid most of the statistical issues present in

<sup>10</sup> We also report the total MAPE as  $(|a| + \frac{1}{N} \sum |\xi_i|)$ , which includes the cross-sectional intercept as a pricing error.

<sup>11</sup> Specifically,  $\text{cov}(\xi) = \frac{1}{T}(I_N - \beta(\beta'\beta)^{-1}\beta')\Sigma_\epsilon(I_N - \beta(\beta'\beta)^{-1}\beta')(1 + \lambda'\Sigma_f^{-1}\lambda)$ , where  $\Sigma_f$  is the variance-covariance matrix of the factors and  $\Sigma_\epsilon$  is the variance-covariance matrix of the time-series errors,  $\epsilon_{i,t}$ .

asset pricing tests that can mechanically produce high explanatory power. Indeed, our simulations show that the probability of a random “noise” factor replicating our results is zero. Importantly, we also show that the model succeeds beyond the highly correlated size and book-to-market portfolios: since the three Fama-French factors explain almost all time-series variation in these returns, the 25 portfolios essentially have only three degrees of freedom. As Lewellen, Nagel, and Shanken (2010) point out, pricing this cross-section with multiple factors is a relatively low hurdle. Below, we show that our one-factor model prices the cross-section of portfolios sorted on size and book-to-market as well as the Fama-French three-factor model. In addition, we avoid the pitfall of relying only on this cross-section by including the more challenging momentum portfolios as test assets. We also show strong pricing performance across U.S. Treasury bond portfolios of various maturities, which further strengthens our results since the model should apply to *all* assets, yet most existing tests only focus on stocks. Finally, the economic motivation of our factor provides further support since it implies that the price of risk should be significant and positive.

We test specifications of the linear factor model (9) in the cross-section of asset returns. As test assets, we consider the size and book-to-market portfolios as well as momentum portfolios, each of which are known to exhibit large cross-sectional dispersion in average returns. We also consider the cross-section of bond returns, using returns on Treasury portfolios sorted by maturity as test assets. We compare our single leverage factor ( $\mathbf{f} = LevFac$ ) to standard benchmark factor models, such as the Fama-French (1993) model ( $\mathbf{f} = [R_{mkt}, R_{SMB}, R_{HML}]$ ), where the comparison benchmark depends on the cross-section of test assets under consideration. We obtain factor and return data from Kenneth French’s data library and the Federal Reserve Board’s Data Releases. Data on equity returns and U.S. Treasury returns are obtained from Kenneth French’s data library and CRSP, respectively. We express all returns and our leverage factor in percent per year (quarterly percent multiplied by four). Our main sample period is 1968Q1 to 2009Q4, but we also report results for the subsample that excludes the recent financial crisis. The results over the precrisis subsample, 1968Q1 to 2005Q4, are marginally weaker than those for the full sample, which suggests that the financial crisis was an important event in revealing the inherent riskiness of some assets.

### III. Main Empirical Results

#### A. Cross-Sectional Analysis

Table III presents our main results. We test the leverage factor model in the cross-section of 41 test assets *simultaneously*. The test assets are: 25 size and book-to-market portfolios, 10 momentum portfolios, and 6 Treasury bond portfolios sorted by maturity. Panel A presents the cross-sectional prices of risk, while Panel B presents several test diagnostics for each model. As comparisons, we consider the CAPM, the Fama-French model, and multifactor models that include the momentum factor as well as the level factor (PC1),

Table III

**Pricing the Size, Book-to-Market, Momentum, and Bond Portfolios**

This table presents pricing results for the 25 size and book-to-market, 10 momentum, and 6 Treasury bond portfolios sorted by maturity. Each model is estimated as  $E[R^e] = \lambda_0 + \beta_{\text{fac}} \lambda_{\text{fac}}$ . FF denotes the Fama-French three factors, Mom the momentum factor, PC1 the first principal component of the yield curve, and LevFac our leverage factor. Panel A reports the prices of risk with Fama-MacBeth and Shanken  $t$ -statistics. Panel B reports test diagnostics, including mean absolute pricing errors (MAPEs) by portfolio group, adjusted  $R^2$ s with corresponding confidence intervals (C.I.), and a  $\chi^2$  statistic that tests whether the pricing errors are jointly zero.  $E[R^e]$  gives the average excess return to be explained. Data are quarterly, 1968Q1 to 2009Q4. Returns and risk premia are reported in percent per year (quarterly percentages multiplied by four).

| Panel A: Prices of Risk   |          |              |              |              |              |           |           |
|---------------------------|----------|--------------|--------------|--------------|--------------|-----------|-----------|
|                           | CAPM     | FF           | FF, Mom      | FF, Mom, PC1 | LevFac       | LevMkt    |           |
| Intercept                 | 3.39     | 3.16         | 1.06         | 0.66         | 0.12         | -0.19     |           |
| $t$ -FM                   | 3.55     | 4.09         | 1.51         | 1.14         | 0.06         | -0.21     |           |
| $t$ -Shanken              | 3.54     | 4.03         | 1.34         | 1.01         | 0.04         | -0.14     |           |
| LevFac                    |          |              |              |              | 62.21        | 60.97     |           |
| $t$ -FM                   |          |              |              |              | 4.62         | 5.29      |           |
| $t$ -Shanken              |          |              |              |              | 3.12         | 3.65      |           |
| Mkt                       | 3.06     | 2.30         | 4.54         | 4.89         |              | 5.46      |           |
| $t$ -FM                   | 0.99     | 0.80         | 1.59         | 1.71         |              | 1.75      |           |
| $t$ -Shanken              | 0.99     | 0.80         | 1.58         | 1.70         |              | 1.55      |           |
| SMB                       |          | 1.76         | 1.57         | 1.63         |              |           |           |
| $t$ -FM                   |          | 0.93         | 0.83         | 0.87         |              |           |           |
| $t$ -Shanken              |          | 0.93         | 0.82         | 0.86         |              |           |           |
| HML                       |          | 3.33         | 4.37         | 4.34         |              |           |           |
| $t$ -FM                   |          | 1.45         | 1.90         | 1.89         |              |           |           |
| $t$ -Shanken              |          | 1.45         | 1.86         | 1.85         |              |           |           |
| MOM                       |          |              | 7.82         | 7.75         |              |           |           |
| $t$ -FM                   |          |              | 2.94         | 2.91         |              |           |           |
| $t$ -Shanken              |          |              | 2.92         | 2.89         |              |           |           |
| PC1                       |          |              |              | 14.99        |              |           |           |
| $t$ -FM                   |          |              |              | 1.03         |              |           |           |
| $t$ -Shanken              |          |              |              | 0.93         |              |           |           |
| Panel B: Test Diagnostics |          |              |              |              |              |           |           |
| MAPE                      | $E[R^e]$ | CAPM         | FF           | FF, Mom      | FF, Mom, PC1 | LevFac    | LevMkt    |
| Size B/M                  | 7.86     | 2.62         | 1.81         | 1.05         | 1.01         | 1.16      | 1.11      |
| MOM                       | 5.80     | 3.05         | 3.75         | 1.47         | 1.48         | 1.79      | 1.85      |
| Bond                      | 1.65     | 1.83         | 1.59         | 0.17         | 0.17         | 0.37      | 0.26      |
| Intercept                 |          | 3.39         | 3.16         | 1.06         | 0.66         | 0.12      | 0.12      |
| Total                     | 6.45     | 6.00         | 5.41         | 2.08         | 1.66         | 1.31      | 1.36      |
| Adj $R^2$                 |          | 0.10         | 0.16         | 0.81         | 0.81         | 0.77      | 0.78      |
| C.I. Adj $R^2$            |          | [0.02, 0.30] | [0.02, 0.36] | [0.74, 0.88] | [0.72, 0.88] | [0.82, 1] | [0.76, 1] |
| $T^2(\chi^2_{N-K})$       |          | 174.48       | 167.46       | 111.45       | 110.19       | 67.87     | 68.86     |
| $p$ -value                |          | 0.0%         | 0.0%         | 0.0%         | 0.0%         | 0.3%      | 0.0%      |



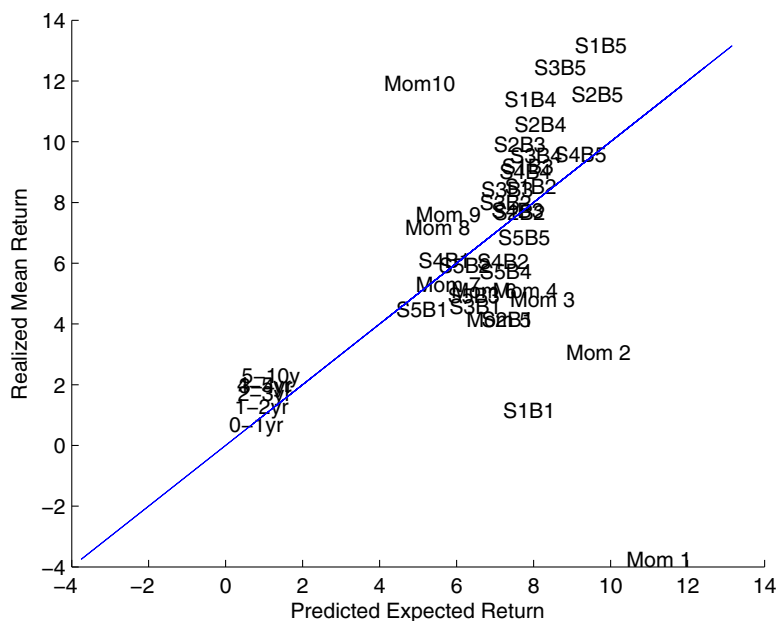
defined as shocks to the first principal component of the yield curve, which prices the cross-section of bond returns (Cochrane and Piazzesi (2009)). These factors constitute the relevant benchmark factors to price the cross-sections considered.

Starting in the first two columns, neither the CAPM nor the Fama-French model is able to account for the spread in average returns across portfolios. Each has a cross-sectional intercept that is economically large at over 3% per annum and statistically significant. The factor prices of risk are not statistically significant and the pricing errors are large, as seen by both the low adjusted  $R^2$  (10% and 16%, respectively) and the  $\chi^2$  test, which measures the sum of squared pricing errors. In Panel B, we also break up the MAPE by asset class. We see that the Fama-French model does relatively well on the size and book-to-market portfolios, with a MAPE of about 2% per annum, but does poorly on the momentum and bond portfolios. The results are substantially better when we add the momentum factor and the level factor. The adjusted  $R^2$  increases to 81%, while the zero-beta rate falls to 66 basis points. The MAPE for each cross-section is fairly low, as is the total MAPE at 1.6% when we include the intercept, which is also a pricing error.<sup>12</sup>

The column marked *LevFac* reports the results for the leverage factor as a sole pricing factor. The cross-sectional intercept is extremely low at 12 basis points and the price of risk is positive and significant. The adjusted  $R^2$  is 77%, while the total MAPE is only 1.3%. In addition, we see that the MAPE for each cross-section is fairly low: 1.2% for the size and book-to-market portfolios, 1.8% for the momentum portfolios, and 0.4% for the bond portfolios. The confidence interval for the  $R^2$  is [82%, 100%], well above the sample value. At first glance, it seems surprising that the lower bound for the confidence interval is *higher* than the sample adjusted  $R^2$ . However, recall that the confidence interval tells us what the most likely values of the *true*  $R^2$  are, *given* the sample adjusted  $R^2$  we observe. Even if the true  $R^2$  were 100%, we would never observe this due to sampling error. Similar logic holds for very high values of the  $R^2$ , and especially with few factors and many test assets where sampling error is larger (see Lewellen, Nagel, and Shanken (2010), Figure 2). Thus, a sample adjusted  $R^2$  of 77% with many assets and a single factor most likely corresponds to a *true*  $R^2$  between 82% and 100% that is biased downward due to sampling error. Finally, the  $\chi^2$  value, while rejected at the 1% level, is still substantially lower (68) than any of the other models (110), despite the fewer degrees of freedom (the statistic associated with the leverage model is  $\chi^2_{N-2}$ , while that associated with the five-factor model is  $\chi^2_{N-6}$ ). In summary, the leverage factor taken on its own does exceptionally well across these portfolios; the performance relative to the five-factor benchmark is quite significant considering the far fewer degrees of freedom.

We plot the predicted versus realized average returns in Figure 1. Aside from the highest momentum portfolio (Mom10), the test assets line up very

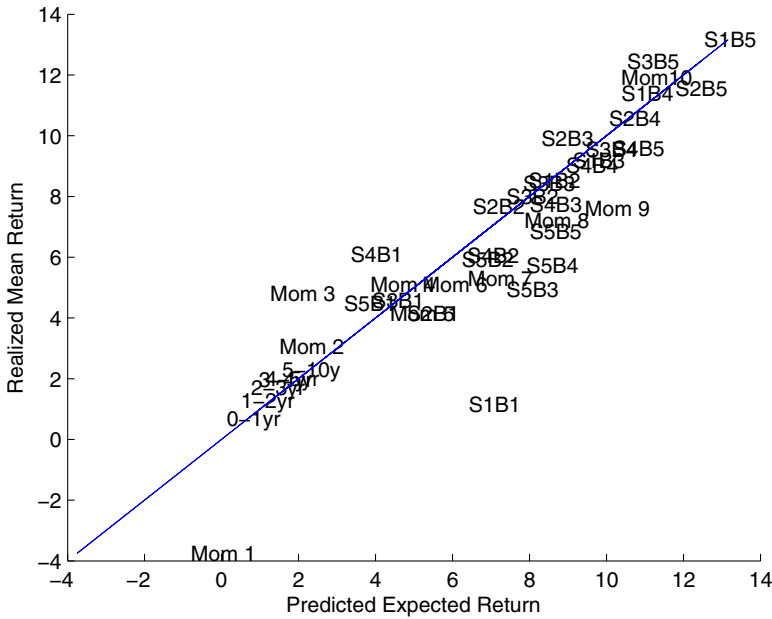
<sup>12</sup> We define the total MAPE as the average absolute pricing error across all 41 portfolios, plus the cross-sectional intercept.



**Figure 4. Realized versus predicted mean returns: Fama-French factors.** We plot the realized mean excess returns of 35 equity portfolios (25 size- and book-to-market-sorted portfolios and 10 momentum-sorted portfolios) and 6 Treasury bond portfolios (sorted by maturity) against the mean excess returns predicted by the Fama-French three-factor benchmark (Mkt, SMB, HML). The sample period is 1968Q1 to 2009Q4. Data are quarterly, but returns are expressed in percent per year.

close to the 45-degree line. We contrast this with the Fama-French model and the five-factor model in Figures 4 and 5. We further examine which portfolios are mispriced in Table IV, which compares the individual pricing errors of the leverage and five-factor models. We notice two patterns: the leverage factor is not able to price the highest momentum portfolio (pricing error of 7%) and neither model does well in pricing the “small growth” portfolio (pricing errors of 5% and 3% for the five-factor and leverage factor models, respectively). Panel B confirms this result by rerunning the cross-sectional tests with each of these portfolios dropped in turn. We see that the explanatory power of the five-factor model increases from 81% to 88% when the small growth portfolio is dropped, and the explanatory power of the leverage factor model increases from 77% to 87% when the highest momentum portfolio is dropped. Importantly, the leverage factor model is no longer rejected if *either* of these portfolios is dropped (the five-factor model remains rejected).

The final column also includes the market return as an additional factor. In the Internet Appendix, we show in detail that this two-factor model is more robust across smaller subsamples. We thus include it for robustness in all of our full-sample tests. However, as we can see, the market return is not statistically



**Figure 5. Realized versus predicted mean returns: five-factor benchmark.** We plot the realized mean excess returns of 35 equity portfolios (25 size- and book-to-market-sorted portfolios and 10 momentum-sorted portfolios) and 6 Treasury bond portfolios (sorted by maturity) against the mean excess returns predicted by a five-factor benchmark model (Mkt, SMB, HML, MOM, and PC1). PC1 represents shocks to the first principal component of the yield curve, which prices the cross-section of bond portfolios. The sample period is 1968Q1 to 2009Q4. Data are quarterly, but returns are expressed in percent per year.

significant and the two-factor model is nearly identical to our single-factor model in terms of all of our test statistics.

It is worth addressing the price of leverage risk, which we estimate to be 62% per annum. While we do not pin down the exact magnitude by theory (only that it should be positive), the number seems economically large. However, note that an inflated price of risk is typical for most nontraded factors because they contain noise that is uncorrelated with returns, which, in turn, tends to deflate the beta estimates. To see this, let

$$LevFac_t = LMP_t + \omega_t,$$

where  $Cov(\omega_t, R_t) = 0$  for any return  $R_t$  and  $LMP_t$  is an LMP—the projection of leverage onto the return space. We return to the LMP in great detail in the next section. Since  $\omega_t$  is noise that is orthogonal to the return space, it will inflate our point estimate of the price of risk. However, since  $\omega_t$  does not affect *covariances*, it will not affect our cross-sectional results in terms of  $R^2$ , etc. Specifically, the presence of  $\omega_t$  will attenuate the time series  $\beta$  of *every* asset by a factor of  $var(LevFac)/var(LMP)$ . It is clear then that the cross-sectional price of risk will have to be higher by exactly this amount to compensate.

**Table IV**  
**Pricing Errors: Pricing the Size, Book-to-Market, Momentum, and Bond Portfolios**

This table presents pricing errors ( $E[R^e] - \lambda_0 - \beta_{\text{fac}}\lambda_{\text{fac}}$ ) in percent per year for our leverage factor model and five-factor benchmark (Mkt, SMB, HML, Mom, and PC1) corresponding directly to the previous table. Panel B reports cross-sectional results when we drop the two most mispriced portfolios: the small growth portfolio (S1B1) and the past winners portfolio (Mom10).

| Panel A: Individual Pricing Errors |                            |                |       |                 |          |        |       |
|------------------------------------|----------------------------|----------------|-------|-----------------|----------|--------|-------|
| Size and Book-to-Market Portfolios |                            |                |       |                 |          |        |       |
|                                    | Low                        | Book-to-Market |       |                 | High     |        |       |
|                                    | $E[R^e]$ : Average Returns |                |       |                 |          |        |       |
| Small                              | 1.15                       | 8.52           | 9.17  | 11.38           | 13.16    |        |       |
|                                    | 4.14                       | 7.67           | 9.91  | 10.55           | 11.54    |        |       |
| Size                               | 4.56                       | 8.00           | 8.43  | 9.54            | 12.46    |        |       |
|                                    | 6.09                       | 6.07           | 7.72  | 9.01            | 9.57     |        |       |
| Big                                | 4.47                       | 5.93           | 4.93  | 5.72            | 6.83     |        |       |
|                                    | Leverage Pricing Errors    |                |       |                 |          |        |       |
| Small                              | −3.27                      | 2.31           | 0.43  | 1.40            | 1.39     |        |       |
|                                    | 0.73                       | −0.42          | 0.46  | −0.10           | −2.10    |        |       |
| Size                               | 0.76                       | 0.79           | 0.09  | 0.49            | 1.88     |        |       |
|                                    | 2.57                       | −1.24          | −0.11 | −0.27           | −0.53    |        |       |
| Big                                | 1.20                       | 1.18           | −0.59 | −2.51           | −2.11    |        |       |
|                                    | Five-Factor Pricing Errors |                |       |                 |          |        |       |
| Small                              | −5.26                      | 0.47           | −0.05 | 0.89            | 0.78     |        |       |
|                                    | −0.64                      | 0.97           | 1.46  | 0.50            | 0.00     |        |       |
| Size                               | 0.52                       | 0.48           | 0.43  | 0.02            | 1.88     |        |       |
|                                    | 2.76                       | −0.43          | −0.28 | 0.03            | −0.45    |        |       |
| Big                                | 1.30                       | −0.18          | −2.32 | −2.14           | −1.06    |        |       |
| Momentum Portfolios                |                            |                |       | Bond Portfolios |          |        |       |
|                                    | $E[R^e]$                   | LevFac         | 5-Fac |                 | $E[R^e]$ | LevFac | 5-Fac |
| Mom 1                              | −3.75                      | −2.46          | −2.96 | 0–1 yr          | 0.70     | 0.15   | −0.17 |
| Mom 2                              | 3.06                       | −1.00          | 1.71  | 1–2 yr          | 1.28     | −0.56  | −0.43 |
| Mom 3                              | 4.80                       | −0.03          | 3.52  | 2–3 yr          | 1.70     | −0.02  | −0.08 |
| Mom 4                              | 5.10                       | −0.63          | 1.02  | 3–4 yr          | 1.95     | −0.27  | −0.03 |
| Mom 5                              | 4.15                       | −2.15          | −0.27 | 4–5 yr          | 2.00     | −0.52  | −0.05 |
| Mom 6                              | 5.11                       | −1.67          | −0.10 | 5–10 yr         | 2.29     | −0.70  | −0.24 |
| Mom 7                              | 5.29                       | −1.23          | −1.23 |                 |          |        |       |
| Mom 8                              | 7.19                       | 0.63           | −0.57 |                 |          |        |       |
| Mom 9                              | 7.61                       | 0.81           | −1.61 |                 |          |        |       |
| Mom 10                             | 11.93                      | 7.25           | 1.81  |                 |          |        |       |

(Continued)

Table IV—Continued

| Panel B: Pricing Results Dropping Largest Pricing Error Portfolios |                |        |                   |        |             |        |           |        |
|--------------------------------------------------------------------|----------------|--------|-------------------|--------|-------------|--------|-----------|--------|
|                                                                    | All Portfolios |        | Drop Small Growth |        | Drop Mom 10 |        | Drop Both |        |
|                                                                    | 5-Fac          | LevFac | 5-Fac             | LevFac | 5-Fac       | LevFac | 5-Fac     | LevFac |
| MAPE                                                               | 1.00           | 1.20   | 0.78              | 1.13   | 0.94        | 1.01   | 0.74      | 0.93   |
| Adj $R^2$                                                          | 0.81           | 0.77   | 0.88              | 0.79   | 0.82        | 0.87   | 0.89      | 0.87   |
| $T^2(\chi^2)$                                                      | 110.19         | 67.87  | 72.42             | 42.95  | 85.15       | 45.78  | 55.27     | 27.40  |
| $p$ -value                                                         | 0.0%           | 0.0%   | 0.0%              | 26.8%  | 0.0%        | 18.1%  | 0.9%      | 87.5%  |

We find this ratio to be about six, making the price of traded leverage risk about 10% per annum—a number much more in line with standard traded factors. Again, it is crucial to understand that the presence of noise like  $\omega_t$  will not impact the cross-sectional results in any way since those rely solely on covariances, but will affect the time-series regressions and cross-sectional price of risk estimates. Specifically, the time-series  $\beta$ 's,  $t$ -statistics, and  $R^2$  will all be deflated (something we return to later) and the cross-sectional price of risk will be inflated.

Having seen the cross-sectional results for all assets simultaneously, we now analyze the cross-sections individually to see more precisely how our leverage factor fares on each set of test assets. Table V reports the results for the 25 size and book-to-market portfolios as well as the 25 size and momentum portfolios. We use the 25 size and momentum portfolios since the 10 momentum portfolios would leave too many degrees of freedom for multifactor models; for example, a four-factor model with an intercept would have five degrees of freedom with 10 portfolios. The results echo and strengthen what we have already seen. For the 25 size and book-to-market portfolios, the leverage factor outperforms the Fama-French factors in terms of the cross-sectional intercept (1% versus 16% per annum), adjusted  $R^2$  (74% versus 68%), and  $p$ -value for the  $\chi^2$  statistic (5.2% versus 0%). The confidence interval for the  $R^2$  is [70%, 100%] and the MAPE is only 2% per annum. The largest absolute pricing error for both models is the well-known “small-growth” portfolio, at 3.7% for the leverage factor and 4.3% for the Fama-French factors. While the model is still rejected at the 10% level, it performs substantially better than the Fama-French factors, which are tailored to explain these portfolios. The price of risk is 56% for the leverage factor, which is close in magnitude to the 62% we estimated in the larger cross-section.

Turning next to the 25 size and momentum portfolios, we compare the leverage factor to the Fama-French and momentum factors. While the adjusted  $R^2$  for the four-factor benchmark is substantially higher (84% versus 51%), the intercept for the four-factor benchmark is substantially higher as well (12% versus 0.3%). The confidence intervals for the  $R^2$  are [72%, 90%] and [40%, 100%], respectively, showing the wide dispersion in  $R^2$ -square values for the leverage factor. Still, the lower bound of 40% is quite high compared to the CAPM. The  $\chi^2$  statistic is fairly low, and the  $p$ -value is 41%, meaning that the model is not rejected. Given the large challenge these portfolios have posed

**Table V**  
**Equity Portfolios**

This table presents pricing results for the 25 size and book-to-market (left panel) and 25 size and momentum portfolios (right panel). Each model is estimated as  $E[R^e] = \lambda_0 + \beta_{fac}\lambda_{fac}$ . FF denotes the Fama-French three factors, Mom the momentum factor, and LevFac our leverage factor. Panel A reports the estimated risk premia, along with Fama-MacBeth and Shanken  $t$ -statistics, respectively. Panel B reports test diagnostics, including MAPEs, the largest absolute pricing error (MAX), adjusted  $R^2$ s with corresponding confidence intervals (C.I.), and a  $\chi^2$  statistic testing whether the pricing errors are jointly zero.  $E[R^e]$  gives the average excess return to be explained. Data are quarterly, 1968Q1 to 2009Q4. Returns and risk premia are reported in percent per year (quarterly percentages multiplied by four).

| Panel A: Prices of Risk               |           |              |           |                                 |           |              |           |           |
|---------------------------------------|-----------|--------------|-----------|---------------------------------|-----------|--------------|-----------|-----------|
| 25 Size and Book-to-Market Portfolios |           |              |           | 25 Size and Momentum Portfolios |           |              |           |           |
|                                       | CAPM      | FF           | LevFac    | LevMkt                          | CAPM      | FF, Mom      | LevFac    | LevMkt    |
| Intercept                             | 12.11     | 15.58        | 1.00      | 0.27                            | 3.51      | 11.72        | 0.31      | -5.47     |
| $t$ -FM                               | 2.99      | 3.84         | 0.25      | 0.06                            | 3.41      | 1.72         | 0.07      | -0.96     |
| $t$ -Shanken                          | 2.97      | 3.57         | 0.18      | 0.04                            | 3.37      | 1.60         | 0.04      | -0.53     |
| LevFac                                |           |              | 55.78     | 56.80                           |           |              | 69.66     | 81.82     |
| $t$ -FM                               |           |              | 3.30      | 3.27                            |           |              | 3.66      | 4.25      |
| $t$ -Shanken                          |           |              | 2.34      | 2.29                            |           |              | 2.28      | 2.38      |
| Mkt                                   | -3.81     | -10.19       |           | 5.20                            | -5.88     | -4.76        |           | 10.62     |
| $t$ -FM                               | -0.80     | -2.09        |           | 0.95                            | -1.17     | -0.64        |           | 1.69      |
| $t$ -Shanken                          | -0.79     | -1.98        |           | 0.70                            | -1.16     | -0.60        |           | 1.00      |
| SMB                                   |           | 1.85         |           |                                 |           | 2.39         |           |           |
| $t$ -FM                               |           | 0.98         |           |                                 |           | 1.12         |           |           |
| $t$ -Shanken                          |           | 0.97         |           |                                 |           | 1.10         |           |           |
| HML                                   |           | 5.76         |           |                                 |           | -4.01        |           |           |
| $t$ -FM                               |           | 2.42         |           |                                 |           | -1.00        |           |           |
| $t$ -Shanken                          |           | 2.38         |           |                                 |           | -0.95        |           |           |
| MOM                                   |           |              |           |                                 |           | 8.40         |           |           |
| $t$ -FM                               |           |              |           |                                 |           | 3.19         |           |           |
| $t$ -Shanken                          |           |              |           |                                 |           | 3.18         |           |           |
| Panel B: Test Diagnostics             |           |              |           |                                 |           |              |           |           |
| 25 Size and Book-to-Market Portfolios |           |              |           | 25 Size and Momentum Portfolios |           |              |           |           |
| MAPE:                                 | CAPM      | FF           | LevFac    | LevMkt                          | CAPM      | FF, Mom      | LevFac    | LevMkt    |
| Intercept                             | 12.11     | 15.58        | 1.00      | 0.27                            | 3.51      | 11.72        | 0.31      | -5.47     |
| Total                                 | 14.41     | 16.69        | 2.09      | 1.34                            | 6.48      | 12.83        | 2.47      | 7.57      |
| MAX                                   | 5.71      | 4.33         | 3.72      | 3.95                            | 9.99      | 4.54         | 7.01      | 6.06      |
| Adj $R^2$                             | 0.03      | 0.68         | 0.74      | 0.75                            | 0.05      | 0.84         | 0.51      | 0.56      |
| C.I.Adj $R^2$                         | [0, 0.28] | [0.48, 0.82] | [0.70, 1] | [0.64, 1]                       | [0, 0.30] | [0.72, 0.90] | [0.40, 1] | [0.38, 1] |
| $T^2(\chi^2_{N-K})$                   | 71.99     | 55.38        | 34.98     | 33.37                           | 75.83     | 50.70        | 23.88     | 18.90     |
| $p$ -value                            | 0.0%      | 0.0%         | 5.2%      | 5.7%                            | 0.0%      | 0.0%         | 41.1%     | 65.1%     |

Table VI  
Treasury Bonds

This table presents pricing results for the six Treasury bond portfolios sorted by maturity. Each model is estimated as  $E[R^e] = \beta_{fac} \lambda_{fac}$ , without an intercept. FF denotes the Fama-French three factors, Mom the momentum factor, PC1 the first principal component of the yield curve, and LevFac our leverage factor. For the traded factors, we report time series  $\alpha$ s as pricing errors, since there are only six test assets. The last column, LevFac NRE (not reestimated), gives pricing errors for the leverage factor using the price of risk from the larger cross-section of stocks and bonds. Panel B reports the estimated risk premia, along with Fama-MacBeth and Shanken  $t$ -statistics, respectively. Panel C reports test diagnostics, including MAPEs, the largest pricing error (MAX), adjusted  $R^2$ s with corresponding confidence intervals (C.I.), and a  $\chi^2$  statistic testing whether the pricing errors are jointly zero.  $E[R^e]$  gives the average excess return to be explained. Data are quarterly, 1968Q1 to 2009Q4. Returns and prices of risk are reported in percent per year (quarterly percentages multiplied by four).

| Panel A: Pricing Errors   |          |      |        |         |              |           |         |           |
|---------------------------|----------|------|--------|---------|--------------|-----------|---------|-----------|
|                           | $E[R^e]$ | CAPM | FF     | FF, Mom | PC1          | LevFac    | Lev NRE | LevMkt    |
| 0–1 yr                    | 0.70     | 0.65 | 0.61   | 0.58    | 0.36         | 0.33      | 0.27    | 0.31      |
| 1–2 yr                    | 1.28     | 1.16 | 1.08   | 0.84    | 0.31         | 0.28      | 0.10    | −0.18     |
| 2–3 yr                    | 1.70     | 1.54 | 1.47   | 1.04    | 0.22         | 0.12      | −0.15   | 0.21      |
| 3–4 yr                    | 1.95     | 1.77 | 1.73   | 1.13    | 0.11         | −0.05     | −0.40   | 0.10      |
| 4–5 yr                    | 2.00     | 1.84 | 1.85   | 1.03    | −0.13        | −0.19     | −0.58   | −0.01     |
| 5–10 yr                   | 2.29     | 2.01 | 2.11   | 0.96    | −0.27        | −0.03     | −0.44   | −0.02     |
| Panel B: Prices of Risk   |          |      |        |         |              |           |         |           |
|                           | PC1      |      | LevFac |         | Lev NRE      |           | LevMkt  |           |
| LevFac                    |          |      | 52.90  |         | 62.21        |           | 40.14   |           |
| $t$ -FM                   |          |      | 2.28   |         | NA           |           | 2.05    |           |
| $t$ -Shanken              |          |      | 1.65   |         | NA           |           | 1.57    |           |
| PC1                       | 31.52    |      |        |         |              |           | 18.88   |           |
| $t$ -FM                   | 2.27     |      |        |         |              |           | 1.95    |           |
| $t$ -Shanken              | 2.14     |      |        |         |              |           | 1.50    |           |
| Panel C: Test Diagnostics |          |      |        |         |              |           |         |           |
| MAPE:                     | $E[R^e]$ | CAPM | FF     | FF, Mom | PC1          | LevFac    | Lev NRE | LevMkt    |
| Total                     | 1.65     | 1.50 | 1.47   | 0.93    | 0.23         | 0.17      | 0.32    | 0.14      |
| MAX                       | 2.29     | 2.01 | 2.11   | 1.13    | 0.36         | 0.33      | 0.58    | 0.31      |
| Adj $R^2$                 |          |      |        |         | 0.78         | 0.85      |         | 0.89      |
| C.I.Adj $R^2$             |          |      |        |         | [0.28, 0.90] | [0.48, 1] |         | [0.48, 1] |
| $T^2(\chi^2_{N-K})$       |          |      |        |         | 17.96        | 9.10      |         | 9.92      |
| $p$ -value                |          |      |        |         | 0.3%         | 10.5%     |         | 4.1%      |

in the literature, we take our results as a relative success in explaining a large amount of variation in the average returns of these portfolios.

Finally, we look at the cross-section of Treasury bond portfolios, sorted by maturities in Table VI. We take average maturities of 0–1, 1–2, 2–3, 3–4, 4–5, and 5–10 years, as reported in the CRSP database. We compare the leverage



factor with the level factor, or shocks to the first principal component of the yield curve (PC1), as well as to the standard equity factors discussed before. With fewer assets, we do not estimate an intercept for the nontraded factors, and, for traded factors, we report the time-series alphas (equivalently, we impose the prices of risk to be equal to the factor means). We report individual pricing errors for each portfolio in Panel A. Panel C gives the MAPE for each model. The average absolute portfolio return is 1.65%, yet the multifactor equity models all have MAPEs greater than 0.9%—meaning they do not explain even half of the average returns. In contrast, the level factor (PC1) has a MAPE of only 23 bps, with an adjusted  $R^2$  of 78%.<sup>13</sup> The leverage factor has a MAPE of merely 17 bps and an adjusted  $R^2$  of 85%. The  $p$ -value of the  $\chi^2$  statistic is 10.5%, meaning the leverage factor model is not rejected, whereas the level factor model is. Moreover, at 53%, the price of risk is broadly consistent with earlier estimates. When we do not reestimate the price of risk (that is, we impose a price of risk equal to 62% as in the full cross-section), the MAPE only increases to 32 bps per annum.<sup>14</sup> Thus, the leverage factor does an excellent job explaining the cross-section of bond portfolios, outperforming the standard benchmarks.

### B. Time-Series Analysis

Table VII reports the results for the time-series regressions of returns on the leverage factor. For each cross-section, we report the average returns, betas,  $t$ -statistics, and  $R^2$ s for the time-series regression of each portfolio. Starting with the size and book-to-market portfolios, we see that the average returns increase from low to high book-to-market portfolios, and generally decrease from small to large (a notable exception is the “small growth” portfolio, which only offers 1% per annum). The leverage betas typically echo this pattern, increasing from left to right as book-to-market increases, and decreasing from top to bottom as market capitalization increases. The  $t$ -statistics for the betas show the same patterns—for higher book-to-market portfolios whose average returns are large, the leverage betas are significantly different from zero, while for portfolios whose average returns are smaller and closer in magnitude to zero, the leverage betas are not statistically different from zero. There is only one portfolio with an average return above 6% per annum whose leverage beta is not significant at the 10% level, and there are no portfolios with an average return above 10% per annum whose leverage beta is not significant at the 1% level. The next panel shows the time-series  $R^2$ s, which increase from left to right and from top to bottom. The values are typically low, as is common for nontraded factors.

<sup>13</sup> In this case, without an intercept, we define the  $R^2$  as  $1 - \frac{(\sum \xi_i^2)}{(\sum (\mu_{R,i} - \bar{\mu}_{R,i})^2)}$  for the model  $\mu_{R,i} = \lambda_f \beta_{i,f} + \xi_i$ , where  $\mu_{R,i}$  is the average excess return on asset  $i$ , and  $\bar{\mu}_{R,i} = \frac{1}{N} \sum \mu_{R,i}$  is the average mean return across assets.

<sup>14</sup> This is important to check since, with few assets and a relatively small spread in average returns, it may be possible to fit this cross-section with an “unreasonable” price of risk.

Table VII  
Time-Series Regressions

This table presents time-series regressions of excess returns on the leverage factor  $R_{i,t}^e = c_i + \beta_{i,Lev} LevFac_t + \epsilon_{i,t}$ . Data are quarterly, 1968Q1 to 2009Q4. Returns are reported in percent per year (quarterly percentages multiplied by four). Leverage betas are multiplied by 100 for convenience.

| Size and Book-to-Market Portfolios                  |          |                |        |       |                 |          |               |        |       |
|-----------------------------------------------------|----------|----------------|--------|-------|-----------------|----------|---------------|--------|-------|
|                                                     | Low      | Book-to-Market |        |       | High            |          |               |        |       |
| $E[R^e]$ : Average (Annualized) Returns             |          |                |        |       |                 |          |               |        |       |
| Small                                               | 1.15     | 8.52           | 9.17   | 11.38 | 13.16           |          |               |        |       |
|                                                     | 4.14     | 7.67           | 9.91   | 10.55 | 11.54           |          |               |        |       |
| Size                                                | 4.56     | 8.00           | 8.43   | 9.54  | 12.46           |          |               |        |       |
|                                                     | 6.09     | 6.07           | 7.72   | 9.01  | 9.57            |          |               |        |       |
| Big                                                 | 4.47     | 5.93           | 4.93   | 5.72  | 6.83            |          |               |        |       |
| $\beta_{Lev}$ : Leverage Betas ( $\times 10^{-2}$ ) |          |                |        |       |                 |          |               |        |       |
| Small                                               | 6.92     | 9.79           | 13.86  | 15.85 | 18.72           |          |               |        |       |
|                                                     | 5.29     | 12.82          | 15.01  | 16.93 | 21.73           |          |               |        |       |
| Size                                                | 5.92     | 11.41          | 13.22  | 14.35 | 16.81           |          |               |        |       |
|                                                     | 5.48     | 11.55          | 12.41  | 14.72 | 16.04           |          |               |        |       |
| Big                                                 | 5.06     | 7.45           | 8.68   | 13.03 | 14.19           |          |               |        |       |
| $t$ -stats                                          |          |                |        |       |                 |          |               |        |       |
| Small                                               | 0.72     | 1.21           | 1.93   | 2.33  | 2.44            |          |               |        |       |
|                                                     | 0.62     | 1.79           | 2.40   | 2.78  | 3.18            |          |               |        |       |
| Size                                                | 0.76     | 1.81           | 2.32   | 2.53  | 2.67            |          |               |        |       |
|                                                     | 0.78     | 1.93           | 2.23   | 2.70  | 2.55            |          |               |        |       |
| Big                                                 | 0.93     | 1.51           | 1.90   | 2.81  | 2.79            |          |               |        |       |
| $R^2$ : $R$ -square                                 |          |                |        |       |                 |          |               |        |       |
| Small                                               | 0.31%    | 0.87%          | 2.18%  | 3.14% | 3.43%           |          |               |        |       |
|                                                     | 0.23%    | 1.89%          | 3.32%  | 4.43% | 5.73%           |          |               |        |       |
| Size                                                | 0.34%    | 1.91%          | 3.12%  | 3.70% | 4.10%           |          |               |        |       |
|                                                     | 0.36%    | 2.17%          | 2.89%  | 4.17% | 3.74%           |          |               |        |       |
| Big                                                 | 0.52%    | 1.34%          | 2.12%  | 4.52% | 4.46%           |          |               |        |       |
| Momentum Portfolios                                 |          |                |        |       | Bond Portfolios |          |               |        |       |
|                                                     | $E[R^e]$ | $\beta_{Lev}$  | t-stat | $R^2$ |                 | $E[R^e]$ | $\beta_{Lev}$ | t-stat | $R^2$ |
| Mom 1                                               | -3.75    | -2.26          | -0.24  | 0.03% | 0-1yr           | 0.70     | 0.69          | 1.89   | 2.63% |
| Mom 2                                               | 3.06     | 6.32           | 0.85   | 0.43% | 1-2yr           | 1.28     | 1.89          | 1.46   | 2.56% |
| Mom 3                                               | 4.80     | 7.56           | 1.23   | 0.89% | 2-3yr           | 1.70     | 2.98          | 2.04   | 3.30% |
| Mom 4                                               | 5.10     | 9.01           | 1.64   | 1.58% | 3-4yr           | 1.95     | 3.79          | 2.06   | 3.58% |
| Mom 5                                               | 4.15     | 9.94           | 1.98   | 2.30% | 4-5yr           | 2.00     | 4.14          | 2.00   | 3.82% |
| Mom 6                                               | 5.11     | 10.71          | 2.07   | 2.49% | 5-10y           | 2.29     | 4.39          | 1.68   | 3.36% |
| Mom 7                                               | 5.29     | 10.30          | 2.26   | 2.98% |                 |          |               |        |       |
| Mom 8                                               | 7.19     | 10.37          | 2.23   | 2.89% |                 |          |               |        |       |
| Mom 9                                               | 7.61     | 10.73          | 2.09   | 2.55% |                 |          |               |        |       |
| Mom10                                               | 11.93    | 7.33           | 1.08   | 0.69% |                 |          |               |        |       |

We see similar patterns for the momentum and bond portfolios. Betas and  $t$ -statistics typically increase along with average returns. A notable exception is the highest momentum decile (the “past winners” portfolio). The leverage factor beta is too small, and the  $t$ -statistic is only 1.08. This is consistent with Figure 1, which graphically shows that this is the most mispriced portfolio for the leverage factor. The bond portfolios typically have larger  $t$ -statistics and  $R^2$ s.

The apparently low  $R^2$ s are again consistent with noise or other uncorrelated measurement error in the leverage factor, as noted above. We do not propose to explain all the movements in leverage over our sample period, which may occur for a number of reasons unrelated to the intermediary SDF. The presence of such noise will lower the  $t$ -statistics and the  $R^2$ s in time-series regressions, but it will not change our pricing results since it does not affect return covariances.<sup>15</sup> The literature often worries about the significance of the time-series betas since one would like to see statistically significant exposures to the portfolios in question. The argument is that if the betas are not well estimated they may be spuriously explaining the cross-section of returns. Our results speak clearly against such spurious relationships: First, we correct for the estimation in betas in our cross-sectional tests and still find statistically significant prices of risk. Second, and most importantly, if one believes that the betas are indeed spuriously explaining the average returns, one needs to evaluate its likelihood, that is, the probability of betas lining up in just the right way to explain the large spread in average returns across 41 assets. In the next subsection, we provide simulation evidence showing that the odds of this happening are essentially zero.

### C. Additional Robustness Tests

To demonstrate the robustness of our results, and to highlight their strength, we show that they are almost certainly not due to chance. Specifically, we simulate a noisy factor by randomly drawing from the empirical distribution of the leverage factor with replacement. We construct this noise factor to have the same length as our original leverage factor (168 quarters) and use it in our cross-sectional pricing tests. Clearly, since this factor is drawn at random, it should not have any explanatory power in the cross-section of expected returns. We repeat this exercise 100,000 times, and ask how likely it is that a “random” factor would perform as well as our leverage factor in a cross-sectional test. The results in Table VIII show that the probability of randomly achieving an  $R^2$  as high as we do, a MAPE as low as we do, and an intercept as low as we do with the observed leverage factor are 0.01%, 0.00%, and 0.19%, respectively.

<sup>15</sup> It can easily be shown that return covariances are unaffected by uncorrelated noise by adding noise to the market portfolio and running repeated cross-sectional pricing tests. While the time-series results can look as noisy as one wants, the cross-sectional results remain unchanged on average as the noise has no covariance with returns.

Table VIII  
Robustness

Panel A shows robustness of our main measure to several starting dates. End dates are 2009Q4 except the last column, which excludes the financial crisis, ending in 2005Q4. Panel B examines how likely a “noise” factor could produce our results. We run 100,000 simulations where we draw randomly from the empirical distribution of our leverage factor. We rerun all our tests, including constructing the LMP (leverage mimicking portfolio). For each statistic, we report the probability that the random factor does as well as our leverage factor (i.e., the probability the random factor has an  $R^2$  as large as our leverage factor or an intercept as small as our leverage factor, or both jointly in the “Joint” column).

| Panel A: Robustness to Various Start Dates |       |       |       |       |       |       |       |              |
|--------------------------------------------|-------|-------|-------|-------|-------|-------|-------|--------------|
| Years                                      | 1966  | 1967  | 1968  | 1969  | 1970  | 1971  | 1972  | 1968 to 2005 |
| Intercept                                  | 0.46  | 0.37  | 0.12  | 0.19  | 0.67  | 1.53  | 1.29  | −0.37        |
| Adj $R^2$                                  | 0.72  | 0.74  | 0.77  | 0.77  | 0.79  | 0.76  | 0.79  | 0.67         |
| $T^2(\chi^2)$                              | 60.49 | 64.20 | 67.87 | 72.00 | 74.72 | 69.07 | 61.83 | 67.61        |
| $p$ -value                                 | 2%    | 1%    | 0%    | 0%    | 0%    | 0%    | 1%    | 0%           |

| Panel B: What Are the Odds a Random Factor Could Produce Our Results? |            |       |       |           |                     |
|-----------------------------------------------------------------------|------------|-------|-------|-----------|---------------------|
|                                                                       |            | $R^2$ | MAPE  | Intercept | Joint $R^2$ -Int    |
| Noise Factor                                                          | $p$ -value | 0.01% | 0.00% | 0.19%     | 0.00%               |
|                                                                       |            | $R^2$ | MAPE  | Sharpe    | Joint $R^2$ -Sharpe |
| Noise LMP                                                             | $p$ -value | 0.18% | 0.00% | 1.63%     | 0.05%               |

Taking these together, there is essentially no chance that the low pricing errors we see are due to chance.<sup>16</sup>

Table VIII also reports on results using different starting dates. We show that the results are robust to starting in the years 1966 to 1972. Similarly, the results in the last column show that the leverage factor does well in the precrisis sample (1968 to 2005), with the  $R^2$  dropping from 77% to 67% and the intercept staying low at 37 bps per annum. We argue that the better performance using the full sample supports our interpretation of the leverage factor—we would expect an intermediary-based factor to reveal stronger risk exposures when we include the financial crisis.

IV. The Leverage Factor–Mimicking Portfolio

To conduct additional tests, we project the leverage factor onto the space of traded returns to form an LMP—a traded portfolio that mimics the leverage factor. This approach has several advantages and provides several new insights. First, since the LMP is a traded return, we can run tests using higher frequency data and a longer time series. This avoids concerns that our

<sup>16</sup> We also construct the “noise” factor using random draws from a normal distribution. The results are very similar, with the  $p$ -values being essentially zero.

results rely on the post-1968 time period or that our results may not hold at a higher frequency. Indeed, we confirm that our findings hold at a monthly frequency going back to the 1930s. Second, we can run individual time-series alpha tests without having to estimate the cross-sectional price of risk, which is not unambiguously pinned down by the intermediary asset pricing theories discussed earlier. We confirm our strong pricing results using time-series alpha tests, including the ability of our factor to help price the three Fama-French factors and momentum factor. Third, we can take a mean-variance approach to our results (see, for example, Hansen and Jagannathan (1991)). We find that the LMP has the largest Sharpe ratio of the traded factors considered and it is close to the maximum possible Sharpe ratio using any combination of the three Fama-French factors and momentum factor. Finally, we use the entire cross-section of CRSP stock returns to construct portfolios based on real-time leverage factor betas and find substantial dispersion in average returns that line up well with the postformation leverage betas.

#### A. Construction of the LMP

To construct the LMP, we project our nontraded broker-dealer leverage factor onto the space of excess returns. Specifically, we run the following regression:

$$LevShock_t = a + b'[BL, BM, BH, SL, SM, SH, Mom]_t + \epsilon_t, \quad (10)$$

where  $[BL, BM, BH, SL, SM, SH, Mom]$  are the excess returns of the six Fama-French benchmark portfolios on size (Small and Big) and book-to market (Low, Medium, and High) in excess of the risk-free rate and  $Mom$  is the momentum factor. We choose these returns for their well-known ability to summarize a large amount of return space: ideally, the error  $\epsilon_t$  is orthogonal to the space of returns so that the covariance of any asset with leverage shocks is identical to its covariance with the LMP, defined as the fitted value of the regression.<sup>17</sup> Most of our results are strengthened if we also include a long maturity bond portfolio. However, our bond data are only available starting in 1952 and one of our main goals is to use a longer history to provide out-of-sample evidence for our results. We thus do not include a bond portfolio in the results. We normalize the weights,  $b'$ , to sum to one, which is without loss of generality and is more convenient in terms of units. The factor mimicking portfolio return is given by

$$LMP_t = \tilde{b}'[BL, BM, BH, SL, SM, SH, Mom]_t,$$

where we estimate  $\tilde{b} = \frac{b}{\Sigma b} = [-0.25, -0.11, 0.62, -0.64, 1.37, -0.48, 0.47]$  via ordinary least squares over 1968 to 2009. While the LMP loads positively

<sup>17</sup> Note that, by construction,  $\text{Cov}(LevShock_t, R_t^e) = \text{Cov}(LMP_t, R_t^e) + \text{Cov}(\epsilon_t, R_t^e) = \text{Cov}(LMP_t, R_t^e)$ , for all  $R_t^e \in \text{span}\{BV, BN, BG, SV, SN, SG, Mom\}_t$ . Since the benchmark factors span a large amount of return space, the covariance of a return with the LMP is expected to be close to its covariance with leverage. However, we acknowledge that some information may be lost in this procedure.

on momentum, the loadings on the other factors are less clear. On net, we do see a higher loading on value as opposed to growth. However, there is no substantial difference between small and large loadings. We find that the correlation between our original leverage factor and the LMP is 0.37.

### *B. Pricing Results Using the LMP*

We investigate the pricing performance of the LMP using the stock and bond portfolios of the previous section as test assets. As before, we begin our tests using quarterly data from 1968 to 2009, but, instead of conducting cross-sectional regressions, we use the time-series alphas for each portfolio. This approach avoids freely choosing the cross-sectional price of risk, since it imposes that the factor risk premium equal the sample mean of the factor return.

For brevity, we report the MAPE by asset class rather than at the individual portfolio level. The results are given in Panel A of Table IX. For comparison, we also report the annualized average absolute return to be explained in the first column. Note that, in this case, the pricing error is simply the time-series alpha. We also report the Gibbons-Ross-Shanken (1986) *F*-statistic (GRS) that tests whether the alphas are jointly zero. We find that the LMP has an average annual alpha of 1.19%, compared with 2.24% and 1.13% using the Fama-French and Fama-French plus momentum factors, respectively. The LMP MAPE is small relative to the 6.33% return to be explained, and nearly as small as the four-factor benchmark. In terms of portfolios, we see that the LMP does well on the size and book-to-market portfolios and especially the bond portfolios (MAPE of 1.15% and 0.59%). On the momentum portfolios, the LMP MAPE of 1.66% is low relative to the Fama-French MAPE of 4.36%, and is fairly close to the four-factor MAPE of 1.46%. The GRS *F*-statistic is 2.57 for the LMP, 2.28 for the four-factor model, and 4.48 for the Fama-French model, though each model is decisively rejected.

Panel B presents cross-sectional results using the LMP and gives important out-of-sample evidence. Using the constant weights we estimate, we compute cross-sectional tests with the LMP based on monthly data going back to 1936.<sup>18</sup> The cross-sectional  $R^2$  for the LMP is 63% versus 81% for the four-factor model and 53% for the Fama-French model. The cross-sectional intercepts are 3%, 15%, and 28% per annum, respectively. Thus, the LMP performs well in comparison to the benchmarks in terms of a relatively high  $R^2$  and much lower intercept. The performance is particularly impressive since the weights used are constant and the LMP cannot by definition outperform the four-factor

<sup>18</sup> While the underlying data are available starting in 1926, we find that none of the benchmark factor models work well including this period. Since the LMP is projected onto these underlying benchmark factors, it cannot perform any better than the factors themselves. Therefore, including the Great Depression does not lend itself to a meaningful comparison, since all models do relatively poorly. Accordingly, we skip the first 10 years of data and start in 1936. The  $R^2$  from the four-factor benchmark model nearly doubles from 46% to 81% going from the 1926 to 1936 starting dates. We also do not use bonds as test assets since bond data are not available in this time period.

**Table IX**  
**The LMP: Comparing Models**

This table presents the time-series alphas generated by each model (LMP, Fama-French, and Fama-French plus momentum). MAPE represents the mean absolute pricing error given in percent per annum ( $\text{MAPE} = \frac{1}{N} \sum_{i=1}^N |\alpha_i|$ ). SBM represents the 25 size and book-to-market portfolios, MOM the 10 momentum portfolios, and Bond the six bond portfolios. The first column (MEAN) gives the absolute average return to be explained. We also report the GRS  $F$ -statistic that the alphas are jointly zero and its associated  $p$ -value. The sample period is 1968Q1 to 2009Q4. The last panel compares cross-sectional results using each of the factor models. We confirm our pricing results using monthly data going back to 1936. For the 1936 to 2009 sample, we only use the size, book-to-market, and momentum portfolios, since bond data are not available.

| Panel A: Time-Series Alphas                             |           |       |         |       |
|---------------------------------------------------------|-----------|-------|---------|-------|
| MAPE                                                    | Mean      | LMP   | FF, MOM | FF    |
| SBM                                                     | 7.86      | 1.15  | 1.04    | 1.57  |
| MOM                                                     | 5.80      | 1.66  | 1.46    | 4.36  |
| Bond                                                    | 3.04      | 0.59  | 0.93    | 1.47  |
| Total                                                   | 6.33      | 1.19  | 1.13    | 2.24  |
| Model Fit                                               |           | LMP   | FF, MOM | FF    |
| GRS                                                     |           | 2.57  | 2.28    | 4.48  |
| $p$ -value                                              |           | 0     | 0       | 0     |
| Panel B: Cross-Sectional Results<br>across Time Periods |           |       |         |       |
| Time Period                                             |           | LMP   | FF, MOM | FF    |
| 1968 to 2009,                                           | Intercept | -0.32 | 1.06    | 3.12  |
| Quarterly                                               | Adj $R^2$ | 0.78  | 0.81    | 0.16  |
| 1936 to 2009,                                           | Intercept | -3.00 | 14.74   | 27.97 |
| Monthly                                                 | Adj $R^2$ | 0.63  | 0.81    | 0.52  |

benchmark on an  $R^2$  basis. Our empirical results thus continue to hold over the longer time span and at the higher monthly frequency, providing out-of-sample robustness tests of our results. Overall, the LMP alone performs comparably to the Fama-French and momentum factors.

Lastly, we compute time-series alphas on the Fama-French and momentum factors themselves, examining whether the LMP can price these portfolios. We use both the main sample period (1968 to 2009) and the earlier sample (1936 to 2009) and report these results in Table X. In the main sample, the time-series alphas are all relatively low, with the exception of momentum, which has an alpha of 4% per annum (about half of the 8% average momentum return) but is not statistically significant. In the earlier sample, the momentum alpha is larger at 5% per annum and is statistically significant, while the alphas on the Fama-French portfolios are all under 1% and are not statistically significant.

One may be concerned that, as the LMP is simply a linear combination of portfolio returns with strong pricing abilities, the mimicking portfolio will also



Table X  
Pricing the Benchmark Factors with the LMP

This table presents time-series regressions of the four benchmark factors, market, SMB, HML, and momentum, on the LMP. We report the quarterly results for our main sample 1968 to 2009, as well as monthly results that begin in 1936. We provide the average mean return to be explained, along with the time-series regression statistics. Mean returns and alphas are reported in annual percentage terms for consistency.

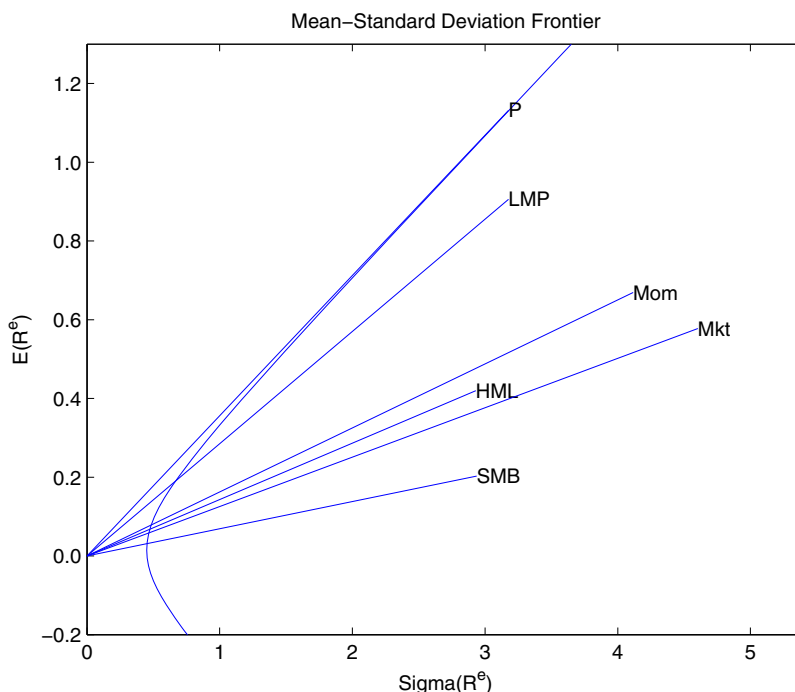
| Model: $R_t^e = \alpha + \beta LMP_t + \varepsilon_t$ |          |       |         |      |         |       |
|-------------------------------------------------------|----------|-------|---------|------|---------|-------|
| Quarterly Data: 1968 to 2009                          |          |       |         |      |         |       |
|                                                       | $E[R^e]$ | Alpha | T-Alpha | Beta | T-Beta  | $R^2$ |
| Mkt                                                   | 5.44     | 0.08  | (0.03)  | 0.52 | (2.90)  | 0.10  |
| SMB                                                   | 3.07     | 1.72  | (0.75)  | 0.13 | (1.19)  | 0.02  |
| HML                                                   | 4.06     | -0.22 | (-0.74) | 0.44 | (2.92)  | 0.12  |
| Mom                                                   | 7.99     | 4.24  | (1.08)  | 0.36 | (1.85)  | 0.06  |
| Monthly Data: 1936 to 2009                            |          |       |         |      |         |       |
|                                                       | $E[R^e]$ | Alpha | T-Alpha | Beta | T-Beta  | $R^2$ |
| Mkt                                                   | 6.93     | -0.22 | (-0.90) | 0.81 | (20.17) | 0.30  |
| SMB                                                   | 2.44     | 0.89  | (0.72)  | 0.16 | (5.25)  | 0.03  |
| HML                                                   | 5.04     | 0.66  | (0.02)  | 0.47 | (17.87) | 0.30  |
| Mom                                                   | 8.03     | 5.08  | (2.86)  | 0.29 | (6.82)  | 0.05  |

mechanically inherit this ability. We show that this is not the case using two approaches. First, we compare the LMP to benchmark models. We find that the LMP *alone* prices nearly as well as the three Fama-French factors and the momentum factor combined. Second, using the simulation technique discussed earlier, we generate a “random” LMP by drawing from the empirical distribution of our leverage factor (with replacement) and projecting the resulting random factor onto the benchmark portfolios. We do this 100,000 times and report the probability that the resulting portfolio is able to replicate the LMP’s pricing ability (see Table VIII). The likelihood of obtaining an  $R^2$  as high as we see is 0.2%, a MAPE as low as we see is 0%, and a Sharpe ratio as high as we see is 1.6%. Thus, it is extremely unlikely that the LMP is spuriously picking up the pricing information of the benchmarks.

C. Mean-Variance Properties of the LMP

We now turn to the mean-variance properties of our mimicking portfolio. Figure 6 plots the efficient frontier implied by the six Fama-French benchmark portfolios, the Fama-French three factors, and the momentum factor. We display the location of each benchmark factor in this space, as well as the line connecting each factor to the origin. Note that the slopes of the lines give the Sharpe ratios of the factors. Recall that a traded return is on the efficient frontier if and only if it is the projection of the SDF onto the return space, which follows from the Hansen and Jagannathan (1991) bounds.





**Figure 6. Risk-return for LMP and benchmark pricing factors.** The hyperbola plots the sample mean-standard deviation frontier (“the efficient frontier”) for the six Fama-French benchmark portfolios, the Fama-French factors (Mkt, SMB, and HML), and the momentum factor. LMP is the leverage mimicking portfolio (see text for description), Mkt, SMB, and HML are the Fama-French factors, Mom is the momentum factor, and P is the linear combination of Mkt, SMB, HML, and Mom that produces the highest possible Sharpe ratio in sample,  $P = \max_{a,b,c,d} [\text{Sharpe}[a\text{Mkt}+b\text{SMB}+c\text{HML}+d\text{Mom}]]$ . Data are monthly from 1936 to 2009.

Figure 6 also plots the portfolio that gives the largest possible Sharpe ratio (0.35) of any linear combination of the three Fama-French factors and momentum factor, labeled  $P$ . Note that, at 0.29, the Sharpe ratio of the LMP is much higher than those of the market (0.13), SMB (0.05), HML (0.15), or even momentum (0.20) factor (see Table XI). In fact, the LMPs Sharpe ratio is comparable to the Sharpe ratio of  $P$ . The annualized (monthly multiplied by  $\sqrt{12}$ ) LMP Sharpe ratio is about one, compared to the maximum possible Sharpe ratio of 1.2. Since the true mean-standard deviation space will in general be much tighter than its sample counterpart, the close proximity of the LMP to the sample mean-standard deviation frontier is an impressive additional piece of evidence that our leverage factor provides a good approximation to the SDF.

#### D. Leverage Beta Sorting

Finally, we follow Fama and French (1993) and form portfolios based on preranking betas, where betas are computed using past 10-year rolling window

Table XI  
Mean-Standard Deviation Analysis

This table presents the monthly mean, standard deviation, and Sharpe ratios of the Fama-French three factors, momentum factor, LMP, and maximum possible Sharpe ratio from any combination of the Fama-French three factors and momentum factor. We provide the annual Sharpe ratio by multiplying by  $\sqrt{12}$ . Data are monthly from 1936Q1 to 2009Q4.

|            | $E[R^e]$ | $\sigma[R^e]$ | Sharpe Ratio | Annualized Sharpe |
|------------|----------|---------------|--------------|-------------------|
| Market     | 0.57     | 4.30          | 0.13         | 0.46              |
| SMB        | 0.15     | 2.86          | 0.05         | 0.18              |
| HML        | 0.40     | 2.75          | 0.15         | 0.50              |
| Mom        | 1.32     | 6.48          | 0.20         | 0.70              |
| LMP        | 1.92     | 3.23          | 0.29         | 0.99              |
| Max Sharpe |          |               | 0.35         | 1.20              |

regressions. We form portfolios based on ex-ante leverage betas and show that the resulting portfolios have a large spread in average returns. Specifically, at each quarter (the formation date), we sort the universe of Amex, NASDAQ, and NYSE stocks from CRSP based on their estimated leverage betas over the past 10 years (40 quarters), which we call preranking betas. We compute leverage betas by regressing quarterly firm-level excess returns (over the three-month risk-free rate) on the leverage factor. Since the leverage factor is constructed in real time, this procedure would have been available to an investor in real time as well. We drop the smallest 10% of firms at formation date to avoid having thinly traded, illiquid firms bias our results. Other filters, such as dropping stocks with a share price under \$5, produce qualitatively similar results. We require at least five years (20 quarters) of nonmissing returns for each stock, making our holding period 1973 to 2009. We group stocks based on preranking betas and form value-weighted portfolios, though our results are qualitatively unchanged using equal-weighted portfolios. We do this each quarter and report the average returns and postformation leverage betas of each group, as well as the average return and beta of a high-minus-low portfolio spread.

Our results, presented in Table XII, Panel A, can most easily be seen by forming three portfolios based on preranking betas (using, for example, five portfolios produces stronger but qualitatively similar results, as shown in Panel B). Using three groups also makes the results directly comparable to the procedure used to construct the Fama-French factors. The leverage betas, average returns, and Sharpe ratios increase mechanically over the low, medium, and high preranked leverage beta portfolios. The resulting high-minus-low portfolio has an annualized spread of 2.8%, which is roughly comparable to that of the SMB and HML factors (3.6% and 4.4% in this sample, respectively). Moreover, the leverage high-minus-low portfolio has a positive and significant leverage beta, with a  $t$ -statistic of 2.01. The CAPM alpha is 2.6% per annum, which is nearly as large as the average return of 2.8%. However, the CAPM alpha is not statistically significant, suggesting that, while there is a spread in returns that is not explained by the CAPM, the portfolio construction is still fairly noisy.

**Table XII**  
**Leverage Beta Sorts**

This table presents sorts of the entire universe of CRSP stocks into portfolios based on preranking leverage betas, found by regressing quarterly firm-level excess returns (over the three-month risk-free rate) on our leverage factor. We use 10 years (40 quarters) for presorting regressions. We exclude the smallest 10% of firms based on market capitalization on formation date and exclude firms that do not have at least 20 quarters of past return data on formation date. We report the mean, standard deviation, Sharpe ratio, postformation leverage beta, and  $t$ -statistic as well as the CAPM  $\alpha$  and associated  $t$ -statistic. We report numbers in percent per year. Leverage betas are multiplied by 100 for convenience. Our holding period is 1973 to 2009.

| Panel A: Three Sorted Leverage Portfolios |       |       |       |                |
|-------------------------------------------|-------|-------|-------|----------------|
|                                           | Low   | Med   | High  | High-Minus-Low |
| $E[R]$                                    | 5.17  | 5.94  | 7.99  | 2.82           |
| $\sigma[R]$                               | 19.63 | 16.99 | 21.01 | 13.44          |
| $E[R]/\sigma[R]$                          | 0.26  | 0.35  | 0.38  | 0.21           |
| Leverage $\beta$                          | 3.81  | 7.76  | 11.39 | 7.58           |
| $t-\beta$                                 | 0.61  | 1.54  | 1.86  | 2.01           |
| CAPM $\alpha$                             | -0.53 | 1.08  | 2.12  | 2.64           |
| $t-\alpha$                                | -0.56 | 1.38  | 1.48  | 1.17           |

| Panel B: Five Sorted Leverage Portfolios |         |       |       |       |          |       |
|------------------------------------------|---------|-------|-------|-------|----------|-------|
|                                          | 1 (Low) | 2     | 3     | 4     | 5 (High) | 5-1   |
| $E[R]$                                   | 4.91    | 5.62  | 5.81  | 7.12  | 8.78     | 3.87  |
| $\sigma[R]$                              | 22.76   | 17.80 | 17.14 | 18.73 | 24.06    | 18.56 |
| $E[R]/\sigma[R]$                         | 0.22    | 0.32  | 0.34  | 0.38  | 0.36     | 0.21  |
| Leverage $\beta$                         | 0.86    | 6.02  | 7.52  | 10.91 | 10.61    | 9.74  |
| $t-\beta$                                | 0.12    | 1.07  | 1.50  | 2.01  | 1.52     | 1.83  |
| CAPM $\alpha$                            | -1.31   | 0.35  | 0.76  | 1.82  | 2.17     | 3.48  |
| $t-\alpha$                               | -0.79   | 0.42  | 0.93  | 1.51  | 1.20     | 1.13  |

The high-minus-low leverage portfolio does not, however, replace the leverage factor in our empirical tests because sorting on nontraded quarterly factor covariances is a noisy procedure. It is well known that sorting on characteristics is less noisy than sorting on covariances, making factors formed on covariance sorts less equipped to capture the underlying discount factor variation. In other words, even if past covariances are perfectly measured, they may not measure *future* conditional covariances well, and in particular characteristics often give a better proxy. This problem is much worse in our case, however, since our factor is nontraded and our data are only of quarterly frequency, meaning estimated covariances with returns are even noisier. The evidence is consistent with such noise: while the correlation between our nontraded leverage factor and the LMP formed earlier is 0.37, the correlation with the high-minus-low leverage beta portfolio is only 0.18. We therefore take the evidence that the

high-minus-low leverage beta portfolio produces a positive average return as additional evidence in support of our model.<sup>19</sup>

In sum, the evidence from the cross-section of individual stock returns lends additional support to our finding that a higher covariance with broker-dealer leverage is associated with higher expected return. Using the entire universe of traded stocks avoids the criticism that leverage betas only explain the average returns across the portfolios we analyze, providing an important additional diagnostic despite the many challenges associated with the procedure.

## V. Discussion of Results

### A. Challenges and Directions for Theory

While we have demonstrated that our leverage factor possesses strong pricing ability across a wide variety of assets and asset classes, we have not provided a formal model that links leverage risk exposure to expected asset returns. The theories reviewed in Section I are broadly consistent with our results, but our empirical findings pose challenges to most of these theories. Of course, any theoretical model, when taken literally, cannot match *every* aspect of the data. Yet, it may be helpful to see where the limitations of the existing theories lie in light of our empirical findings. Thus, much like the theories that guided our empirical tests in linking financial intermediaries to asset prices, we now hope that our empirical results will help guide future theoretical work in grappling with the facts. We analyze each theory, in turn, and discuss the potential clashes.

We noted that broker-dealer leverage may measure the tightness of borrowing constraints or funding liquidity (Brunnermeier and Pedersen (2009)). This interpretation gives rise to procyclical leverage (“the margin spiral”) and is potentially an important source of macroeconomic risk. However, our findings present two challenges to the mechanics of the margin spiral. First, we find that leverage shocks are largely uncorrelated with shocks to the Pástor-Stambaugh (2003) liquidity factor—a measure of innovations to market liquidity—challenging the theoretical predictions that funding liquidity and market liquidity are intertwined.<sup>20</sup> Second, our results hold well across different time periods, including both good times and crises. It seems less likely that broker-dealers are borrowing-constrained when times are good, yet we pick up important risk exposures over these periods also.

Broker-dealer leverage may also be a signal of the wealth of the financial system, consistent with models where the return on financial sector wealth determines expected returns (He and Krishnamurthy (2013)). As the wealth

<sup>19</sup> In results reported in the Internet Appendix, we also find that sorting stocks based on LMP betas provides even more support. Using the LMP allows us to use monthly data and a longer time series (1936 to 2009) to compute more accurate betas. The spread in returns in this case is economically larger. However, since the LMP is constructed using the full sample, the strategy is not implementable in real time for an investor, in contrast to the strategy employed above.

<sup>20</sup> Results are available upon request.

of the financial sector increases, demand for services from broker-dealers increases, leading them to leverage up capital. However, this intuition assumes that broker-dealer leverage is never constrained, which is inconsistent with our understanding of the events during the financial crisis (Gorton and Metrick (2012)). Moreover, direct measures of financial sector wealth, such as the value-weighted equity return of financial institutions, do not perform as well as leverage in explaining the cross-section of average returns. We relegate these results to the Internet Appendix. However, we also note that, in He and Krishnamurthy (2013), it is the bank manager's wealth that is the SDF and this can be difficult to measure.

A further challenge to the theories by He and Krishnamurthy (2013) and Brunnermeier and Sannikov (2014) is our finding of a positive price of risk of intermediary leverage. The positive price of leverage risk reflects the procyclicality of leverage. The theories of Brunnermeier-Sannikov (2014) and He-Krishnamurthy (2013), on the other hand, feature countercyclical intermediary leverage.

The theory that matches our empirical findings most closely is that by Adrian and Boyarchenko (2012). Their theory gives rise to a pricing kernel with shocks to intermediary leverage. In addition, the price of risk of intermediary leverage is always positive, a finding that we confirm across assets classes and time periods. Adrian and Boyarchenko's (2012) pricing kernel has aggregate output as an additional risk factor, with a price of risk that generically fluctuates between positive and negative. While we do not control for aggregate output, we do control for the market return in our baseline regressions and find an insignificant price of risk for the market, though we confirm in the Internet Appendix that including the market in the regressions increases the stability of the leverage price of risk estimates over time. Notably, the finding of an insignificant price of risk for the market return is consistent with Adrian and Boyarchenko's (2012) model. Their model also gives rise to predictions about the evolution of the price of risk over time. These time-series predictions are further explored by Adrian, Moench, and Shin (2013). In particular, the market price of risk relates negatively to the (detrended) level of intermediary leverage. This is the time-series analog to the procyclicality of leverage: a run-up in leverage pushes up asset prices, compressing expected returns, which corresponds to a low market price of risk.

### *B. Further Economic Interpretation of Funding Constraints*

We have argued based on theory that leverage may measure funding constraints faced by the financial sector. Here, we provide additional supporting evidence that our leverage factor measures funding constraints. As Frazzini and Pedersen (2013) show, investors who face funding constraints will tend to prefer high-beta assets that do not require leverage, thus bidding up their price and creating a spread between high- and low-beta assets. To see this, consider the portfolio choice of an investor who can hold either a high-beta stock or a low-beta bond combined with an instantaneous risk-free asset—for

simplicity, assume that the investor can hold only *one* of the risky assets and has no other wealth. Modern portfolio theory guides the investor to hold the asset with the highest Sharpe ratio, combining it with risk-free borrowing or lending, depending on risk aversion. However, if the investor is sufficiently risk tolerant and cannot take leverage or is funding-constrained, she will prefer to hold the stock because it provides a higher expected return without the use of leverage. Therefore, a spread arises between the leveraged bond and the stock, which comoves with the tightness of funding conditions—as funding constraints tighten, investors bid up the prices of riskier assets relative to safer assets. Using this logic, Frazzini and Pedersen (2013) argue that stocks sorted on market betas should display a spread in returns that comoves with funding conditions. If our leverage factor measures funding constraints, this spread should also be related to our leverage factor.

To construct beta-sorted portfolios, we follow the procedure in Fama and French (1992). Specifically, in June of each year, we sort the universe of Amex, NASDAQ, and NYSE stocks from CRSP based on past market betas and form 10 portfolios. We use 10-year betas (120 months) and require 60 months of observations for a firm to be included. This is in contrast to Fama and French (1992), who use five-year betas (60 months) and require 24 months of data. We find that using the longer history for the estimation of betas produces a wider spread in returns and a substantially greater Sharpe ratio for the low-minus-high portfolio.<sup>21</sup> As before, we drop the lowest decile of firms based on market capitalization on formation date to avoid illiquid, thinly traded firms whose returns may be unreliable, although this is not crucial for our results. We then compute excess returns on the sorted portfolios over the following year. We rescale each of these portfolios to have a market beta of one. That is, we leverage up the low-beta securities and combine the high-beta securities with the risk-free asset in such a way that the ex-post market betas of all 10 portfolios are one. Note that, since these are excess returns, multiplying each portfolio by a constant still gives us excess returns and hence these are valid test assets. We find results consistent with the previous literature: the annual spread between a leveraged low-beta security and a high-beta security (low-minus-high) is about 7% per annum (see Table XIII, Panel A). We also see average returns and Sharpe ratios that are decreasing from low to high.

Consistent with our interpretation of the leverage factor, the correlation between the leverage factor and the low-minus-high beta portfolio is positive, 0.22. That is, as funding tightens and leverage decreases, low-beta stocks that require high leverage underperform high-beta stocks. The correlation is sizable given that the correlation between the leverage factor and the LMP is 0.37. This finding complements the results in Table II where we show that the leverage factor is correlated with variables commonly associated with funding constraints (volatility, Baa-Aaa spread, and asset growth). Note that the leveraged low-beta portfolios all have large and statistically significant leverage

<sup>21</sup> Our results are weaker but qualitatively similar when we use sorts based on five-year betas.

**Table XIII**  
**“Betting against Beta” Portfolios**

This table presents sorts of CRSP stocks on 10-year preranking market betas. We lever up low-beta stocks so that each portfolio has a market beta of one. The resulting portfolios comove with funding conditions since leverage-constrained investors will bid up high-beta securities (Frazzini and Pedersen (2013)). See the text for a complete description. FF represents the Fama-French factors, and Mom adds the momentum factor. The holding period is 1968 to 2009. All data are quarterly but returns and Sharpe ratios are presented as annualized numbers (multiplied by 4 and 2, respectively).

| Panel A: Time-Series Regressions: $R_{i,t}^e = c_i + \beta_{Lev,i} LevFac_t + \epsilon_{i,t}$ |          |        |                           |           |       |
|-----------------------------------------------------------------------------------------------|----------|--------|---------------------------|-----------|-------|
|                                                                                               | $E[R^e]$ | Sharpe | $\beta_{Lev} (x 10^{-2})$ | $t$ -stat | $R^2$ |
| BAB1                                                                                          | 10.98    | 0.46   | 19.45                     | 2.93      | 4.90% |
| BAB2                                                                                          | 8.94     | 0.40   | 21.71                     | 3.50      | 6.88% |
| BAB3                                                                                          | 7.29     | 0.36   | 16.41                     | 2.91      | 4.84% |
| BAB4                                                                                          | 6.87     | 0.35   | 11.33                     | 2.01      | 2.38% |
| BAB5                                                                                          | 6.68     | 0.34   | 11.67                     | 2.11      | 2.60% |
| BAB6                                                                                          | 4.67     | 0.25   | 12.91                     | 2.41      | 3.38% |
| BAB7                                                                                          | 5.68     | 0.30   | 10.19                     | 1.89      | 2.10% |
| BAB8                                                                                          | 4.68     | 0.25   | 8.90                      | 1.67      | 1.66% |
| BAB9                                                                                          | 4.29     | 0.22   | 3.97                      | 0.72      | 0.31% |
| BAB10                                                                                         | 3.99     | 0.20   | 3.51                      | 0.62      | 0.23% |
| 1 – 10                                                                                        | 6.99     | 0.36   | 15.94                     | 2.90      | 4.82% |

| Panel B: Cross-Sectional Test Diagnostics |          |      |         |           |                |
|-------------------------------------------|----------|------|---------|-----------|----------------|
| MAPE:                                     | $E[R^e]$ | FF   | FF, Mom | LevFac    | LevFac, No Int |
| Intercept                                 |          |      |         | 2.48      |                |
| Total                                     | 6.41     | 1.36 | 0.61    | 3.29      | 1.33           |
| MAX                                       | 10.98    | 4.06 | 1.81    | 2.14      | 2.32           |
| Adj $R^2$                                 |          |      |         | 0.73      | 0.51           |
| C.I. Adj $R^2$                            |          |      |         | [0.42, 1] | [0.20, 1]      |
| $T^2(\chi^2)$                             |          |      |         | 7.38      | 6.62           |
| $p$ -value                                |          |      |         | 49.61%    | 67.65%         |

factor betas and relatively high time-series  $R^2$  values, all of which decrease as we move toward the high-beta portfolios.

We next test our leverage factor in the cross-section of the beta-sorted portfolios. The results are presented in Panel B of Table XIII. Consistent with our previous results, the leverage model produces a high cross-sectional  $R^2$  of 73% and the cross-sectional price of risk associated with the leverage factor is positive and significant. However, at 33% the price of risk is relatively low compared to our earlier estimates. When we estimate the model without an intercept, the price of risk increases to 50%, more in line with our earlier estimates, but the  $R^2$  falls to 51%. In either case, the model is not rejected. We also compare the pricing error of the leverage model with that of the Fama-French three-factor model and Fama-French-Carhart four-factor model. The one-factor leverage model with no intercept has an average absolute pricing error of 1.33%,



which compares with 1.36% and 0.61% for the three- and four-factor models, respectively.

Note that we choose to exclude the beta-sorted portfolios from our main test portfolios because they are largely subsumed by standard risk factors for U.S. equities (the four-factor alphas are essentially zero and the four-factor time-series  $R^2$  are high). In other words, they add no additional degrees of freedom to our main test assets. This is in line with Kogan and Papanikolaou (2012), who show that the variation in beta-sorted portfolios is related to the variation in book-to-market-sorted portfolios and a common principal component explains both. However, the beta-sorted portfolios do serve to enhance the economic interpretation of our leverage factor as one related to funding constraints.

## VI. Conclusion

In this paper, we focus on measuring the SDF of a representative financial intermediary using the aggregate leverage of security broker-dealers. Our approach is motivated by a growing theoretical literature in which the pricing kernel is a function of intermediary balance sheet variables. Our results are most closely related to models in which intermediary leverage comoves with funding constraints. When funding constraints tighten, intermediaries are forced to deleverage by selling off assets, and these are times when their marginal value of wealth is high. Since financial intermediaries trade in many markets, have low transactions costs, optimize frequently, and use extensive models to make investment decisions, an SDF based on a representative intermediary should have greater empirical success than its conventional counterparts.

Our empirical results are remarkably strong. We show that broker-dealer leverage as the *single* risk factor outperforms the Fama-French model in the cross-section of portfolios sorted on size and book-to-market, outperforms the level factor in the cross-section of bond portfolios, and compares well to the benchmark tailored to explain the cross-section of size- and momentum-sorted portfolios. Furthermore, the leverage factor prices the combined cross-section of all the above portfolios remarkably well. The success of the leverage factor across all these cross-sections is measured in terms of high adjusted  $R^2$  statistics, low values for the  $\chi^2$  statistic based on the sum of squared pricing errors, small and statistically insignificant cross-sectional pricing errors, and cross-sectional prices of risk that are statistically significant and consistent across portfolios. When taking all these criteria into account, our single factor performs as well as standard four- and five-factor models tailored to price the cross-sections considered, and provides an economic explanation for their average returns. We perform a battery of additional tests that confirm the robustness of our results.

Our study is a first step in exploring how the marginal value of wealth of intermediaries can be used as a pricing kernel. We see the search for additional measures of the intermediary SDF as a fruitful area for further research. We also regard empirical tests that distinguish between competing theories, and



ultimately a cohesive theory that can quantitatively match the empirical facts, as particularly promising areas for future work.

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### Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

**Appendix S1:** Internet Appendix.